RESEARCH CONTRIBUTION FOR PROVEIT

A fixed-base, modular Diophantine equation for ZFC consistency

Reusing a binomial modulus as the proof-label base

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Main delivered equation. An explicitly specified polynomial in **230 unrestricted integer variables**, of exact degree four, defined by 176 quadratic residuals. Its collected expansion has 4,482 nonzero monomials and coefficient height 7,968.

$$P(\boldsymbol{z}) = \sum_{i=0}^{175} R_i(\boldsymbol{z})^2 = 0.$$

The ordinary proof, composed with the retained numerical-calculus calibration, establishes

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{z} \in \mathbb{Z}^{230}\ P(\boldsymbol{z}) = 0\right) \iff \mathrm{Con}(\mathrm{ZFC}).$$

Every residual is printed in Appendix A and supplied as exact machine-readable data. No unspecified coefficient or variable exponent remains in the delivered polynomial.

Evidence boundary. This is an ordinary mathematical proof with reproducible finite algebraic checks. It is not a new end-to-end Lean/Rocq formalization, a proof of consistency, or a claim of global minimality.

Abstract

We simplify the direct proof-certificate route to a Diophantine equation for syntactic consistency of ZFC. The central change is to give one binomial coefficient two jobs. With a proof horizon t, set $R = 16^t$ and $B = \binom{R+t}{t}$. The number B supplies both the prime-power selectors for proof rows and the evaluation point in the label weights $1 + jB$. An elementary bound for $\mathrm{lcm}(1, \ldots, t)$ proves the required row coverage, while $B > (t+1)^t$ proves exact label-multiplicity recovery. No certificate of factorial authenticity, and no separate variable-base growth power, is needed.

We then compose this arithmetic front with the predecessor's modular numerator method in the current signed Pell architecture. The new compatibility argument allows all nine CRT representatives to remain signed, recovers the necessary positive quantities from existing power requests, and moves eleven numerator outputs out of the magnitude frame. A new cofinal two-square completion of the Pell center restores the common modular-root congruence. Only six exponential requests remain exact; successor denominators use multiplication. The resulting finite quadratic system is contracted with explicit integral restoration maps and converted to the literal quartic displayed in this report. The contribution is a structural improvement within the direct, small-coefficient family; few-variable constructions with different coefficient and degree costs remain distinct alternatives.

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1 The exact result and its scope

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$ and quantify every coordinate of the principal equation over all of $\mathbb{Z}$. The file `equations/balanced/equation.json` declares the ordered variables and every quadratic residual. A sparse term $[c, [i_1, \ldots, i_d]]$ denotes $cz_{i_1} \cdots z_{i_d}$; an empty index list denotes the constant c. The finite sum of residual squares is the definition of P. The separate `expanded_polynomial.json` gives every collected coefficient.

Theorem 1.1 (Consistency equivalence)**.** *For the supplied fixed polynomial $P \in \mathbb{Z}[z_0, \ldots, z_{229}]$, ZFC proves that an integer zero exists if and only if a finite syntactic ZFC refutation exists. Consequently ZFC proves*

$$\neg\exists \boldsymbol{z} \in \mathbb{Z}^{230}\ P(\boldsymbol{z}) = 0 \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

The exact degree of P is four.

The companion in `equations/compact/equation.json` uses 229 integer variables and 175 quadratic residuals. It has 12,006 collected monomials and the same coefficient height 7,968. It is obtained by one additional integral affine elimination, so it has an integer zero exactly when P does. Appendix B prints its full residual list as well. The principal version is preferred when the shorter expansion matters; the companion is preferred when one fewer coordinate matters more.

The proof does not assume either consistency or inconsistency. In particular, proving the equivalence does not prove the absence of zeros. The syntactic consistency predicate is the one calibrated by the repository's fixed numerical calculus C_{222}, with truth label 0 and falsity label 1. The retained logical interfaces are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Their source and provenance accompany the package; their complete transitive proof-assistant development was not rebuilt here.

The new arithmetic arguments below are finite number theory and primitive recursive coding arguments formalizable in ZFC. Their composition with these inherited calibrations proves Theorem 1.1. A fresh machine-checked object-language ZFC derivation is a separate, unfinished task. Exact coefficient checks verify the finite artifacts and their algebraic transformations, not the unbounded semantic theorem by themselves.

1.1 What "simpler" means in this report

There is no single total ordering of variable count, degree, coefficient height, monomial count, arithmetic circuit cost, and proof length. The repository already contains equations with far fewer unknowns, including quartics with large fixed coefficient payloads. This report optimizes the direct proof-calculus family, which retains modest literal coefficients and avoids both a $K215$ simulation and an instantiation of Jones's universal equation.

The comparison snapshot is the immutable ProveIt commit

`b7e35224c04fe1031c0a53f80a395e9b32205022`.

We inspected its active research documents, delivered packages through 147, and all eight incoming archives present at that commit. The top-level comparison tables lagged behind the newest incoming material; the comparison below therefore uses the actual incoming endpoints.

Profile	Variables	Rows	Degree	Monomials	Height
Matched parent	252	182	4	5,340	128,446
Factorial-free, sparse	242	178	4	5,018	7,968
Factorial-free, compact	241	177	4	12,774	7,968
Shared error, minimum	240	177	4	21,482	6,144
Shared error, lower height	241	178	4	9,544	4,512
Normalized, balanced	244	176	4	5,212	3,968
New principal	**230**	**176**	**4**	**4,482**	**7,968**
New compact	**229**	**175**	**4**	**12,006**	**7,968**

Every row in this table quantifies over unrestricted integers. Height means the largest absolute coefficient after collecting the expanded quartic. The source paths and exact metric files for the comparison rows are in `provenance/comparison.json`.

The modular numerator device itself is inherited: the retained `INHERITED_MODULAR_FACTORIAL_PROOF.md` already uses six exact powers, eleven modular numerators, and a common Pell-root congruence. What is changed here is its compatibility with the fixed-base/binomial weight front and the later signed, pooled certificate architecture. The dual use of B, the new sign bootstrap, the two-square cofinal-center theorem, and their complete composition were not found in the inspected snapshot. No literature-wide priority claim is made.

2 The arithmetic design before polynomial compilation

The following notation explains certified register values. Powers and binomial coefficients in this section are not extra operations in the final quartic. They are replaced by the polynomial certificates discussed below and printed literally in the appendix.

Choose a cofinal proof horizon $t = 16 + h^2$. The essential front is

$$R = 16^t, \qquad B = 2b + 1, \qquad B = \binom{R+t}{t}, \tag{1}$$

$$y = RB(2v+1)^2, \qquad H = 2 + 2(R+t) + C^2 + \sum_{j=1}^{8} A_j^2, \tag{2}$$

$$N = B^x, \qquad q = (BH)^x = 6z + 3. \tag{3}$$

Here $C, A_1, \ldots, A_8$ are the actual CRT representatives. Some raw stored coordinates equal these values minus one; every literal source edit preserves that distinction. The nine representatives are unrestricted integers. Their small residues recovered later are nonnegative.

The two beta endpoints and the row-index congruence are

$$x = B + 1 + (y+1)u_0, \tag{4}$$

$$x = 1 + [1 + (t+1)y]u_t, \tag{5}$$

$$Bh_B = C + R + 1. \tag{6}$$

Every Horn weight is now $1 + jB$. The former independent growth base $F = [2(t+1)]^t$ and the radius F^2 are removed from the semantics. The same B is used throughout; changing only some occurrences would not give the claimed construction.

The six exact requests are

$$R = 16^t, \quad N = B^x, \quad q = (BH)^x, \quad D_t = q^t, \quad D_y = q^{y+1}, \quad M = q^x. \tag{7}$$

Set $D_{t+1} = qD_t$ by multiplication. The eleven binomial numerator indices are $R+t$, C, the eight representatives A_j, and the derived label $J(\boldsymbol{A})$. Their relevant lower indices are $t, t+1, y+1, x$. The exact-binomial digit for B uses t; the index range uses t; four syntax ranges and the derived-label range use $t+1$; two accumulator ranges use $y+1$; two quotient ranges use x.

3 A fixed-base radius and a binomial with two roles

3.1 The radius supplies every row selector

Let $\Lambda_t = \operatorname{lcm}(1, \dots, t)$, with $t \geq 1$.

Lemma 3.1. *For every $n \geq 1$, $\Lambda_n \leq 4^{n-1}$. In particular, $16^t \geq \Lambda_t^2$.*

Proof. Let $m = \lceil n/2 \rceil$. The quotient Λ_n/Λ_m divides $\binom{n}{\lfloor n/2 \rfloor}$. To see this, its valuation at any prime is at most one: there is at most one new prime power between $m+1$ and n. If such a prime power occurs, its contribution to the factorial valuation of the central binomial coefficient is one, and all other contributions are nonnegative.

For $n = 2r$, induction and the binomial row sum give

$$\Lambda_{2r} \leq \Lambda_r \binom{2r}{r} \leq 4^{r-1} 2^{2r} = 4^{2r-1}.$$

For $n = 2r+1$, the two equal central coefficients together are at most the row sum, so $\binom{2r+1}{r} \leq 2^{2r}$. Hence

$$\Lambda_{2r+1} \leq \Lambda_{r+1} \binom{2r+1}{r} \leq 4^r 4^r = 4^{2r}.$$

The initial case $n = 1$ is immediate. The stated radius consequence is weaker than the resulting strict bound and follows at once. □

The next selector theorem is inherited from the incoming factorial-free binomial CRT report, and is restated with its proof because it is the key interface used by the new fixed radius. The classical lcm/binomial divisibility used in its proof is also related to Farhi's identity [3].

Lemma 3.2 (Coverage and compatibility of selectors)**.** *Let $t \geq 1$, $R \geq \Lambda_t^2$, and $B = \binom{R+t}{t}$. For every $i \in \{1, \dots, t\}$ there is a full prime power $p^e \| B$ such that $p^e \mid R+i$ and $p^e > t$. Conversely, every full prime power $p^e \| B$ divides at least one of $R+1, \dots, R+t$.*

Proof. For the first assertion, $R + i > \Lambda_t^2$, so some prime p has $v = v_p(R+i) > 2a$, where $a = \lfloor \log_p t \rfloor$. Since $p^v > t$, the valuation of $R+j$ for $j \neq i$ is exactly $v_p(j-i)$. Therefore

$$e = v_p(B) = v - \delta, \qquad \delta = v_p\left(t \binom{t-1}{i-1}\right).$$

For each p^r, the corresponding factorial-valuation summand for δ is zero or one, and vanishes when $p^r > t$. Thus $0 \leq \delta \leq a$, so $a < e \leq v$ as required.

For the converse, put $v = \max_i v_p(R+i)$. The valuation of B is the sum over $r \geq 1$ of

$$\left\lfloor \frac{R+t}{p^r} \right\rfloor - \left\lfloor \frac{R}{p^r} \right\rfloor - \left\lfloor \frac{t}{p^r} \right\rfloor.$$

Each term is zero or one and all terms above $r = v$ vanish. Hence $e \leq v$, and a factor attaining the maximum is compatible. □

The two directions have different uses. Coverage proves soundness for each row. Compatibility supplies assignments at every prime of the global modulus in the CRT completeness construction, including primes not chosen as a particular row's selector.

3.2 The same binomial separates label multiplicities

Lemma 3.3. *For $t \geq 1$ and $R = 16^t$,*

$$B = \binom{R+t}{t} > (t+1)^t.$$

Moreover B is odd.

Proof. The inequality $16^t > t(t+1)$ follows by induction: its left side grows by a factor of 16 and the right side by at most 3. Every factor $(R+i)/i$, $1 \leq i \leq t$, is greater than $t+1$, proving the first claim by the product formula for B.

For oddness, $v_2(R) = 4t > \lfloor \log_2 t \rfloor$. Consequently $v_2(R+i) = v_2(i)$ for every $1 \leq i \leq t$. The valuations in the numerator product and in $t!$ cancel exactly. □

Here is the large-evaluation argument inherited from the factorial-free predecessor, now applied to B. Suppose the decoded t proof rows have conclusion labels c_i and premise labels p_i, q_i in $[0, t]$, and the accumulator updates imply

$$\prod_{i<t}(1 + c_i B) = (1+B)\prod_{i<t}(1 + p_i B)(1 + q_i B). \tag{8}$$

Discard factors with label zero. If the two sides have different numbers of remaining factors, the smaller factor count is at most t, while $B > (t+1)^t$ makes a product with one more factor strictly larger than any product with the smaller count. Thus the counts agree and are at most t.

Replace B by an indeterminate T. Each resulting product has nonnegative coefficients and coefficient sum at most $(t+1)^t < B$. If their difference were nonzero, let $a_r T^r$ be its lowest nonzero term. The equality at $T = B$ would imply $B \mid a_r$, although $0 < |a_r| < B$. Thus the products agree in $\mathbb{Z}[T]$. Unique factorization, or their distinct rational roots $-1/j$, gives the exact multiplicity balance for each positive label.

To finish the flow argument, let D be the set of labels derivable by the valid Horn rules. It contains truth. A rule whose conclusion is outside D has at least one premise outside D. Summing conclusion counts minus premise counts over the complement of D therefore gives a nonpositive number. If falsity were outside D, the exact balance (8) would give one instead. Hence falsity is derivable. No acyclicity assumption about the chosen multiset of rules is needed.

4 Pell values, the free center, and modular numerators

For an integer A, define the Pell polynomials by

$$X_0 = 1, \quad Y_0 = 0, \qquad X_{k+1} = AX_k + (A^2-1)Y_k, \quad Y_{k+1} = X_k + AY_k,$$

and extend them to negative indices by $X_{-k} = X_k$ and $Y_{-k} = -Y_k$. The retained square frame, semantic anchor, and pooled error certificate recover exactly these pairs at every requested signed index. The complete fourteen-pair orientation pool is retained.

For a genuine exact request with base n, the certified congruence is

$$X_k(A) + (n - A)Y_k(A) \equiv m \pmod{2An - n^2 - 1}.$$

The magnitude frame bounds every exact base, exponent, and output atom in absolute value by W. Its large-center bound makes the modulus larger than the possible error, giving $m = n^k$ for $|n| \geq 2$ and forcing $k \geq 0$. If $k < 0$, the same recurrence would force the impossible integer equality $mn^{-k} = 1$. These signed recovery arguments are inherited from the retained source proofs.

4.1 Why the center must be freed

The current signed frame restricts its abscissa to a particular Pell sequence. This alone does not justify choosing it in an arbitrary congruence class. Retain the lower frame with abscissa L and use

$$A = L^2 + u^2 + v^2. \tag{9}$$

This gives $A \geq L^2$ at every integer solution. Completeness requires a special argument: an arbitrary prescribed nonnegative difference is not a sum of two squares. The frame itself is existentially chosen, and its finite residue behavior provides the missing freedom. The cofinal completion lemma below proves exactly what is needed. It saves one coordinate relative to the initially verified constructor $A = L^2 + u^2 + v^2 + w(w + 1)$.

Only the lower frame uses L; the principal Pell pairs and the semantic anchor use A. The magnitude frame excludes the eleven modular numerator outputs, which now depend on A. It also removes redundant atoms with separately proved bounds. In particular, keeping t, $R + t$, and q controls the fixed base 16, R, $t + 1$, and $q + 1$: $W \geq t^2 + 2 \geq 258$ and

$$R^2 \leq 2((R + t)^2 + t^2) \leq 2(W - 2) < W^2.$$

The successor output D_{t+1} no longer needs an exact-power magnitude bound. Every retained exact request and modular index still has its required strict bound. These changes are part of the literal frame certificate, rather than a numerical estimate on sample inputs.

4.2 A single modular-root equation

Set $\Omega = NM$ and impose

$$2A(q + 1) - (q + 1)^2 - 1 = \Omega w. \tag{10}$$

For every signed integer index n, retain the polynomial numerator

$$\mu_n = X_n(A) + (q + 1 - A)Y_n(A). \tag{11}$$

It need not equal the enormous integer $(q + 1)^n$.

Lemma 4.1 (Signed modular powers)**.** *Under* (10)*, with* $\Omega \neq 0$*,*

$$\mu_n \equiv (q + 1)^n \pmod{\Omega} \qquad (n \in \mathbb{Z}),$$

where a negative exponent means the inverse power in the residue ring.

Proof. Write $s = q + 1$. Equation (10) says $s(2A - s) \equiv 1 \pmod{\Omega}$, so s is a unit. The Pell recurrence and $(s - A)^2 \equiv A^2 - 1$ show that the evaluated Pell unit at index n is s^n. This proves positive indices by induction, and inverse powers give negative indices. □

In completeness, B and q are odd, and $N \mid q$; hence Ω is odd and every prime dividing Ω divides q. The root congruence specifies

$$A_0 = \frac{(q+1)^2+1}{2(q+1)} = 1 + \frac{q^2}{2(q+1)} \pmod{\Omega}.$$

Its denominator is invertible. The following lemma constructs this residue class using precisely the two-square center.

Theorem 4.2 (Cofinal two-square center)**.** *Let $W \geq 2$ and let q, Ω be positive odd integers such that every prime of Ω divides q. There exist a positive odd k and integers u, v such that, with $L = X_{Wk}(W+1)$,*

$$A = L^2 + u^2 + v^2 \equiv A_0 \pmod{\Omega}.$$

The value A can be made arbitrarily large, and $Y_{Wk}(W+1)/W$ is an odd integer. Thus the old odd-quotient frame and the new common root remain simultaneously complete.

Proof. If $\Omega = 1$, choose $k = 1$ and arbitrary sufficiently large u; the frame-parity argument below still applies. Assume $\Omega > 1$. Put $D = W(W+2)$ and work in the finite quadratic ring

$$S_\Omega = (\mathbb{Z}/\Omega\mathbb{Z})[s]/(s^2 - D).$$

The element $e = (W+1) + s$ is a norm-one unit. Write the finite order of e^W as $2^r k$ with k odd. Then $g = e^{Wk}$ has power-of-two order. Its coefficients are $L = X_{Wk}(W+1)$ and $Y = Y_{Wk}(W+1)$.

Fix $p^a \| \Omega$. If $L \equiv \varepsilon \pmod{p}$ with $\varepsilon \in \{1, -1\}$, then the image of g is exactly ε in S_p. When $D \not\equiv 0 \pmod{p}$, this follows from $L^2 - DY^2 = 1$. When $D \equiv 0 \pmod{p}$, the nilpotent part of $\varepsilon + Y s$ has p-power order, and is killed only trivially by a power of two because p is odd. This proves the assertion also in the ramified case.

The kernel of the unit reduction map $S_{p^a}^\times \to S_p^\times$ is a finite p-group: its elements are $1 + pz$. The element εg belongs to this kernel and has power-of-two order, so it is the identity. Consequently $L^2 \equiv 1 \pmod{p^a}$.

Since $A_0 \equiv 1 \pmod{p}$, the difference $A_0 - L^2$ is a unit unless $L \equiv \pm 1 \pmod{p}$. In the latter case it equals $q^2/(2(q+1))$ modulo p^a, and is either zero or has the even valuation $2v_p(q)$. Every residue of zero or even valuation modulo an odd prime power is a sum of two squares. For a unit, the two sets $\{a^2 : a \in \mathbb{F}_p\}$ and $\{c - b^2 : b \in \mathbb{F}_p\}$ intersect for any nonzero residue c, since each has $(p+1)/2$ elements. At least one resulting coordinate is nonzero, so its invertible derivative lifts the solution to every p^a. For valuation $2b$, multiply both coordinates of a unit representation by p^b.

CRT, applied separately to the two square coordinates, therefore gives $u^2 + v^2 \equiv A_0 - L^2 \pmod{\Omega}$. Replacing u by $u + j\Omega$ makes A arbitrarily large.

It remains to check the literal frame parity. The recurrence modulo W gives $W \mid Y_W(W+1)$. If W is odd, the ordinate is odd. If W is even, its base is odd, every abscissa is odd, and $Y_{2m} = 2X_m Y_m$ gives $v_2(Y_W) = v_2(W)$; odd-index ordinates are odd. Hence $Y_W(W+1)/W$ is odd in both cases. Finally,

$$Y_{Wk}(W+1) = Y_W(W+1)Y_k(X_W(W+1)),$$

whose second factor is odd for odd k. Thus the quotient remains odd and supplies the old witness in $L^2 - W^3(W+2)(2z+1)^2 = 1$. □

Remark 4.3 (Why the frame choice matters). This theorem does not work by holding the least frame point fixed. For example, $W = 4$, $q = 3$, $\Omega = 9$ give the least frame abscissa $L = 4801$ and $A_0 - L^2 \equiv 3 \pmod{9}$, which is not a sum of two squares modulo 9. Choosing the odd multiplier from the finite unit order is essential. The proof does not add an order computation or an extra quantified object to the polynomial; it chooses an existing existential frame witness in the completeness direction.

5 Signed binomial digits from modular data

For $n \in \mathbb{Z}$ and $k \in \mathbb{N}$, use the integer-valued generalized binomial coefficient

$$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!}, \qquad \binom{n}{0} = 1.$$

For $n = -a < 0$, this equals $(-1)^k\binom{a+k-1}{k}$.

Lemma 5.1 (Uniform coefficient bound)**.** *For every $n \in \mathbb{Z}$ and $k \geq 0$,*

$$\left|\binom{n}{k}\right| \leq \max(1, |n|)^k.$$

Proof. The ordinary nonnegative case is immediate. For $n = -a < 0$, $\binom{a+k-1}{k}$ counts multisets of size k from a colors. Every multiset has a sorted word, and there are a^k words. The case $k = 0$ is included. □

Lemma 5.2 (The range implication for signed upper indices)**.** *If p is prime, $e, k \geq 1$, and $p^e \mid \binom{n}{k}$ for $n \in \mathbb{Z}$, then $n \bmod p^e < k$.*

Proof. If the falling factorial contains zero, then $0 \leq n < k$, and the claim is immediate. Otherwise its valuation minus $v_p(k!)$ is a sum of terms, each zero or one: a block of k consecutive integers contains either $\lfloor k/p^r \rfloor$ or one more multiples of p^r. If the block contained no multiple of p^e, then $k < p^e$ and every term at level $r \geq e$ would vanish. The valuation would be at most $e - 1$, a contradiction. A multiple in the block gives $n \equiv j \pmod{p^e}$ with $0 \leq j < k$. □

The implication is one-sided. Its converse is not used. Completeness will use higher CRT precision to compensate for division by $k!$.

5.1 Finite truncation, including negative indices

The formal identity $(1+T)^n = \sum_{j\geq 0}\binom{n}{j}T^j$ holds in $\mathbb{Z}[[T]]$ for all signed n, using the formal inverse for $n < 0$. It may equally be proved modulo T^{k+1} by a finite recurrence, so no analytic convergence claim is involved. Suppose $q > 0$, $N \mid q$, $1 \leq k \leq x$, and $D = q^k$. Define

$$r_0 = \sum_{j=0}^{k-1}\binom{n}{j}q^j.$$

The modulus ND divides $\Omega = Nq^x$, and the omitted terms of degree at least $k+1$ vanish modulo ND. Lemma 4.1 therefore gives

$$\mu_n \equiv r_0 + D\binom{n}{k} \pmod{ND}. \tag{12}$$

This remains true at the largest lower index $k = x$ because $N \mid q$.

5.2 The common centered norm

Write $\mathcal{G}(a,b,c) = a^2 + b^2 + c(c+1)$. This is nonnegative on $\mathbb{Z}^3$, and represents every nonnegative integer. Indeed, Legendre's three-square theorem represents $4g+1$ as three squares. Exactly one square is odd modulo four; write the coordinates as $2a, 2b, 2c+1$. Dividing the resulting equality by four gives $g = a^2+b^2+c(c+1)$. This is the inherited ternary gap constructor; it is distinct from the new two-square center theorem.

For each of the eleven digits, let D_i be its already certified denominator and put $D_i v_i = M$. Retain the normalized row

$$v_i \mu_{n_i} = MQ_i + \rho_i. \tag{13}$$

For the exact digit $Q_i = B + qh_i$; for a range digit $Q_i = Nh_i$. For $D_i = M$, $v_i = 1$ is a constant rather than a new coordinate. The single norm budget is

$$M^2 = 1 + 4\sum_{i=1}^{11} \rho_i^2 + \mathcal{G}(\alpha, \beta, \gamma). \tag{14}$$

Since $v_i \mid M$, equation (13) implies $v_i \mid \rho_i$. Thus $r_i = \rho_i / v_i$ is an integer and $\mu_{n_i} = D_i Q_i + r_i$. The norm gives $|r_i| < D_i/2$. If $|r_0| < D_i/32$, then $D_i \mid r_i - r_0$ and $|r_i - r_0| < D_i$, so $r_i = r_0$. Applying (12) to a range digit yields

$$N \mid \binom{n_i}{k_i}.$$

For the exact digit, the stronger modulus $qD_t \mid \Omega$ yields $B \equiv \binom{R+t}{t} \pmod{q}$.

The argument does not require a signed representative's truncation to be a least nonnegative remainder. Its small absolute value is sufficient. Canonical completeness representatives are positive, but soundness applies to every integer zero of the final equation.

6 Soundness in a noncircular order

6.1 Recover positive arithmetic before reading digits

The inherited Pell index and orientation proof applies first. Its magnitude frame now contains the six exact requests and every modular numerator index; it does not contain the modular outputs. The free-center row gives $A > 0$ and the required magnitude separation.

The request with base 16 gives $R = 16^t > 1$. The affine radix condition $q = 6z + 3$ gives $|q| \geq 3$, and the explicitly odd B is nonzero. The exact request $D_y = q^{y+1}$ forces $y + 1 \geq 0$. But $y = RB(2v+1)^2$ is a nonzero multiple of R. A nonpositive y would therefore be at most $-R < -1$. Thus $y > 0$ and $B > 0$.

The possible value $B = 1$ needs separate treatment before binomial authentication. For every signed k, the Pell evaluation at base 1 is congruent to 1 modulo $2(A-1)$: the Pell unit evaluates to the unit 1 in that residue ring. Hence the retained exact request forces $N \equiv 1 \pmod{2(A-1)}$. Its frame bound $|N| < W$ and the large-center bound force $N = 1$. Thus $N = B^x$ holds also in this temporary base-one case. For $B \geq 3$ it follows from the usual exact power lemma once $x \geq 0$ is recovered below.

The square-defined H is positive for arbitrary signed representatives. Hence the request $q = (BH)^x$ gives $x \geq 0$ and $q > 0$. The affine radix excludes $x = 0$. Since $y \geq RB > B$, a negative u_0 in (4) would imply $x \leq B - y < 0$. Therefore $u_0 \geq 0$ and $x \geq B + 1 > 1$. Equation (5) now forces $u_t \geq 1$, giving

$$x \geq 2 + (t+1)y, \qquad y \geq RB > B, \qquad x > 2^{32}. \tag{15}$$

This order uses neither the sign of a CRT representative nor the value of a binomial digit. In particular, it avoids using factoriality or a binomial interpretation to prove the signs needed for that interpretation.

All four denominator groups are now positive divisors of $M = q^x$, and $N = B^x$ is positive and divides q. The local majorant and derived-label budgets can be interpreted at this point.

6.2 Bounds for every relevant coefficient

Set $\ell = 2^{22}$ and $E = 2^{23}$. The retained local arithmetic ledger bounds the derived-label polynomial in absolute value by H^ℓ and bounds the decoded nonnegative local rule polynomial by

$$0 \le \mathcal{L} < (2 + t + x + y + B)^E.$$

The signed derived-label bound has a separate structural audit. For $H \ge 2$, assign exponent 1 to each bounded stored input. At an actual sparse gate $g = \sum_m c_m \prod_{i \in m} v_i$, the auditor assigns

$$e(g) = \left\lceil \log_2 \sum_m |c_m| \right\rceil + \max_m \sum_{i \in m} e(v_i).$$

Induction proves $|g| \le H^{e(g)}$. The supplied ledger checks 91 ancestor gates and only four free leaves. It obtains exponent 3,649,856 for the stored wire $J - 1$, and $3{,}649{,}857 < \ell$ for J. It does not assume positivity of the inputs. The stored inputs are actual representatives minus one; $H \ge 2 + a^2$ implies $|a - 1| \le H$ for every signed integer a.

Before binomial authentication, (15) and $R = 16^t$ already give $x > 2\ell(t + 1)$ and $x > 2t$. For the actual representatives, $\max(1, |A_j|) \le \sqrt{H}$ and likewise for C. Thus Lemma 5.1 bounds every relevant coefficient of an ordinary range by $H^{x/2}$. For the derived label, the same bound follows from $\ell(t+1) < x/2$. For the main binomial, $R + t < H$ and $t < x/2$ give it again. Consequently, for all eleven digits and all relevant positions,

$$\left|\binom{n}{j}\right| \le H^{x/2} \le \sqrt{q}.$$

As q is already far greater than 4096,

$$\frac{|r_0|}{q^k} \le \frac{\sqrt{q}}{q - 1} < \frac{1}{32}.$$

All common-norm reconstructions are therefore valid, including those with negative upper indices.

For the main digit, both B and $\binom{R+t}{t}$ are positive and less than $q/2$: the former follows from $B \le BH$ and $q = (BH)^x$ with $x \ge 3$, and the latter from its square-root bound. Their congruence modulo q is therefore equality. This authenticates $B = \binom{R+t}{t}$.

6.3 Decode rows and recover a finite refutation

The range congruences have the form

$$N \mid \binom{C}{t}, \qquad N \mid \binom{A_j}{b_j + 1}, \qquad N \mid \mathcal{L}(C, \boldsymbol{A}),$$

with an additional range for the polynomially derived label. For row $i - 1$, choose p^e from Lemma 3.2. By Lemma 5.2, the ordinary residue of C modulo p^{xe} lies in $[0, t - 1]$. Equation (6) identifies that residue modulo p^e with $i - 1$. Since $p^e > t$, it is exactly $i - 1$. All other columns recover their bounded nonnegative local values in the same way.

The inequalities in (15) give $2+t+x+y+B<2x$. The elementary inequality $(2x)^{2^{23}}<2^x$ for $x>2^{32}$ then bounds every decoded local value by $2^x\le p^{xe}$. Its divisibility by p^{xe} forces it to vanish. For the derived label, the decoded syntax inputs lie in $[0,t]$. Apply the same structural ledger with input bound $t+1$ to obtain

$$0\le J(\text{local syntax})\le (t+1)^{\ell}<(2+t+x+y+B)^{E}<2^x.$$

Its small range residue and its locally evaluated value are congruent modulo p^{xe} and both lie in $[0,p^{xe})$, so they agree. Every row is valid.

The recovered beta accumulator starts at $B+1$ and ends at 1. The inequality $y>B$ makes the initial remainder proper without a boundary exception. Valid local updates have positive weights $1+jB$; starting from $B+1>0$, all accumulators are positive. Multiply the updates and cancel them to obtain (8). Lemma 3.3 and the flow argument then give a finite C_{222} refutation. The retained logical calibration gives a finite ZFC refutation.

7 Constructive completeness

Start with a finite C_{222} refutation, expand it to a proof tree, and choose $t=16+h^2$ beyond every required row count and label bound. Truth padding supplies exactly t rows. Set $R=16^t$ and $B=\binom{R+t}{t}$. Build the pending accumulators using weights $1+jB$.

Let O be the odd part of $t!$. Choose an arbitrarily large positive odd multiple w of O and set $y=RBw^2$. By Lemma 3.3, B is odd; $v_2(R)=4t>v_2(t!)$. Thus y is an even multiple of $t!$, and can exceed every accumulator. The beta moduli $1+(i+1)y$, $0\le i\le t$, are pairwise coprime: a common prime would divide $j-i\le t$ and hence y, an impossibility. They are all odd. CRT supplies the beta code x; choose a positive, sufficiently large even lift. The endpoint quotients and local beta quotients then satisfy their required bounds. The latter are strictly below x because their moduli are at least two.

Set $N=B^x$, which is odd. For each full prime power $p^e\|B$, use the compatible row from Lemma 3.2, and prescribe the row's actual values modulo p^{2xe}. CRT supplies all nine representatives modulo N^2. The extra precision is essential: division by any range factorial $k!$, with $k\le x$, loses at most $k-1<x\le xe$ powers of p. The resulting binomial therefore remains divisible by p^{xe}. This proves completeness of the range conditions without using a false converse to Lemma 5.2.

Choose the dominant schema representative a odd and the other eight representatives even. The inherited label constructors give $J\equiv 1+a+b \pmod 2$ and $J\ge 2a^2$, so the semantic anchor is even. These parity choices make H odd. If $3\mid B$, then BH is already a multiple of 3. If $3\nmid B$, the modulus N^2 is coprime to 6, and the representatives may additionally be chosen so that $H\equiv 0 \pmod 3$. Two independent square residues suffice: their possible sums modulo 3 are 0, 1, and 2. This adjustment preserves the stated parities and the CRT classes.

Choose the other representatives positive and at most $6N^2$; take a in a compatible class between $2N^2$ and $8N^2$. Then $J\ge 8N^4$ dominates twice every other requested index. The front indices satisfy $R+t<B<N^2$ and $t,y+1,x<N^2$. Increasing the representatives changes the self-sized radix using their actual values, so it does not invalidate the coefficient bounds.

Now $q=(BH)^x$ is a positive odd multiple of 3, hence it has the required affine form $6z+3$. Compute the exact powers in (7), the successor denominator, the actual index atoms, and their magnitude frame. Choose the lower frame abscissa L and a sufficiently large $A=L^2+u^2+v^2$ by Theorem 4.2, so that (10) holds. The inherited anchor and pooled orientation completion at this A supplies all requested Pell pairs. Use their polynomial values (11) as the eleven modular numerators.

The genuine normalized remainders have absolute value below $M/32$. Therefore

$$4\sum_{i=1}^{11}\rho_i^2 < \frac{44}{1024}M^2 < M^2.$$

The remaining nonnegative integer gap in (14) has a $\mathcal{G}$ representation. All quotient witnesses are integral by the common-root and range congruences. This constructs a zero of the new raw polynomial system from every finite refutation, without making any choice depend circularly on the subsequently chosen Pell center.

8 The literal polynomial and finite verification

Every source definition is a quadratic integer equation. The builder records the exact changes from the pinned parent, the recomputed magnitude frame, the full power-request census, and every deleted private coordinate. It then applies the retained Horn factorization and polynomial contractions.

A permitted monic contraction removes a row of the form $\epsilon z_i + f = 0$, where $\epsilon \in \{1, -1\}$ and f does not involve z_i. Substitution gives an integer polynomial inverse, and the checker requires every remaining residual to stay quadratic. Recorded affine consequences and integer residual-row operations preserve the common zero set. For example, replacing R_i by $R_i + cR_j$ is reversed by subtraction. Integer coordinate translations are bijections on the declared domain. The sum-of-squares polynomial may change under these operations, but its zero set remains equivalent.

Finite audit	Count
Literal exact-power requests retained	6
Modular numerator definitions checked	11
Principal signed Pell pairs checked	14
Retained frame atoms	21
Independent Horn factoring edits	18
Integral monic restorations	208
Coordinate translations before final refinements	69
Final balanced / compact affine projections	2 / 3
Deliberately corrupted algebra artifacts rejected	7

Each collected coefficient is checked independently by summing the literal residual squares. The normal and optimized-Python checks give the same algebraic results. The complete packet also reproduces all 41 newly built stage and endpoint JSON files byte for byte. The two-square-center regressions cover 2,210 completions and 3,128 prime-power cases, including the counterexample to keeping an arbitrary frame fixed.

The verification code uses explicit exceptions for failed conditions, so optimized Python does not disable its checks. The arithmetic regressions cover signed generalized binomial coefficients, signed range implications, fixed-base binomial parity and growth, prime-power row selectors, and common-root modular digits. These finite cases are useful error detectors; the proofs in the preceding sections are what establish the unbounded claims.

8.1 Reproduction and integration

From a fresh extraction, with Python 3.10 or later:

```
python3 -B verify.py
python3 -O -B verify.py
```

The verifier is read-only unless a receipt path is explicitly supplied. The archive includes deterministic builders and exact source-to-output certificates; no network access or external symbolic algebra system is needed for verification. To rebuild the article with ordinary TeX Live, run `build_report.sh`. The standalone TeX source contains all generated report inputs. PDF metadata may vary by TeX installation; the exact polynomial data do not.

The contribution can be placed as a new directory in `Computability/DiophantineConsistency/con-zfc-diophantine-forms`. This work does not modify an upstream branch. Source licenses and attribution notices are retained. The integrity manifest is a local checksum inventory, not an external signature or a proof certificate.

9 What remains open

The delivered result is a complete, fixed polynomial with an ordinary consistency-equivalence proof. It does not establish the smallest possible variable count, the smallest coefficient height, or a global optimal arithmetic circuit. Nor does it make a nine-natural-variable or ten-integer-variable universality result obsolete. In particular, Sun's September 2026 ten-integer-variable theorem [4] and the contemporary universal-bound work [5] concern different complexity tradeoffs.

The central reusable result here is the source architecture: one binomial supplies row selectors and multiplicity separation; one existing exponent request supplies the needed sign; and one cofinal center supports all modular numerators. The finite code measures the complete composition. Further improvements should target the remaining six exact powers, the private anchor witnesses, or a different local proof calculus. Merely shortening a displayed abbreviation, omitting a signed case, or discarding a completeness condition would not be an improvement.

References

[1] V. Reshetnikov, *ProveIt*, inspected commit `b7e35224c04fe1031c0a53f80a395e9b32205022`. https://github.com/VladimirReshetnikov/ProveIt. The exact inherited source and archive hashes are in the supplied provenance ledger.

[2] *Removing factorial authenticity from Diophantine consistency*, incoming ProveIt research packet, 25 October 2026. Retained in `upstream/factorial_free/`. The binomial selector and large-evaluation lemmas are inherited from this packet; its earlier modular numerator sources are separately credited.

[3] B. Farhi, *An identity involving the least common multiple of binomial coefficients and its application*, American Mathematical Monthly 116 (2009), 836–839. https://arxiv.org/abs/0906.2295.

[4] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/abs/2609.01594.

[5] J. Bayer, M. David, M. Haßler, Y. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3, 22 October 2026. https://arxiv.org/html/2506.20909v3.

A Complete principal residual list

The only unknowns are $z_0, \dots, z_{229} \in \mathbb{Z}$. Define $P = \sum_{i=0}^{175} R_i^2$ using the following literal list. There are no additional quantified symbols. This list and the declared integer domain specify the entire equation; the JSON file uses the same zero-based coordinate ordering.

$$
\begin{aligned}
R_0 &= z_3 - z_4 - 2z_4\, z_{228} \\
R_1 &= z_0 - z_0\, z_2 - z_2 + z_7 \\
R_2 &= -1 - z_2\, z_5 + z_6\, z_7 - 2z_{228} \\
R_3 &= -1 + z_1 - z_6\, z_7 \\
R_4 &= 1 + z_4 + z_{16} - z_{18} - 2z_{18}\, z_{228} \\
R_5 &= -1 - z_8^2 - 2z_8\, z_9 - z_9^2 + z_{19} \\
R_6 &= -10z_8 + z_8^2 - 6z_8\, z_9 - 8z_8\, z_{19} - 20z_9 - 11z_9^2 - 16z_9\, z_{19} \\
&\quad - 15z_{19} - 4z_{19}^2 + z_{20} \\
R_7 &= -2 + 2z_8 - z_8^2 - 4z_8\, z_{20} - 4z_{20} - 4z_{20}^2 + z_{21} \\
R_8 &= 1 + 6z_8 - 36z_8^2 - 24z_8\, z_{21} + 2z_{21} - 4z_{21}^2 + z_{22} \\
R_9 &= -2 - z_9 - z_9^2 - 2z_9\, z_{10} - z_{10} - z_{10}^2 + z_{23} \\
R_{10} &= -8 + 7z_8 - z_8^2 + 4z_8\, z_{10} - 4z_8\, z_{23} - 4z_{10} + 4z_{23} + z_{24} - z_{46} \\
R_{11} &= -1 + z_8 - z_8^2 - 2z_8\, z_{10} + z_{10} - z_{10}^2 + z_{25} \\
R_{12} &= 23 + 16z_8 - 8z_8\, z_9 - 8z_8\, z_{10} - 16z_8\, z_{19} - 8z_8\, z_{25} + 19z_9 \\
&\quad + 9z_9^2 - 36z_9\, z_{10} - 12z_9\, z_{25} + 16z_{10} + 8z_{10}^2 - 24z_{10}\, z_{19} - 8z_{19}\, z_{25} \\
&\quad - z_{20} + z_{26} - z_{31} \\
R_{13} &= -4z_{24}^2 - 8z_{24}\, z_{26} - 4z_{26}^2 + z_{27} \\
R_{14} &= -2 - 8z_8 + 8z_8\, z_{23} + 8z_8\, z_{25} - 8z_9 + 8z_9\, z_{23} + 8z_9\, z_{25} - 4z_{19} \\
&\quad + 8z_{23} - 4z_{23}^2 - 8z_{23}\, z_{25} + 8z_{25} - 4z_{25}^2 + z_{28} \\
R_{15} &= 6 + 4z_8 + 32z_8\, z_9 + 16z_8\, z_{19} + 16z_8\, z_{25} + 8z_8\, z_{28} + 24z_9 \\
&\quad - 32z_9\, z_{25} - 16z_9\, z_{28} + 12z_{19} - 16z_{19}\, z_{25} - 8z_{19}\, z_{28} - z_{20} + 8z_{25} \\
&\quad - 16z_{25}^2 - 16z_{25}\, z_{28} + 4z_{28} - 4z_{28}^2 + z_{29} \\
R_{16} &= 22 - 5z_8 - z_8^2 - 8z_8\, z_9 - 12z_8\, z_{10} - 4z_8\, z_{23} - 8z_9 - 16z_9\, z_{10} \\
&\quad - 16z_9\, z_{23} - 20z_{10} - 20z_{10}^2 - 24z_{10}\, z_{23} - 28z_{23} - 4z_{23}^2 + z_{30} \\
R_{17} &= 19 - 25z_8 - 7z_8^2 - 8z_8\, z_9 - 30z_8\, z_{10} - 16z_8\, z_{25} + 3z_9 + 5z_9^2 \\
&\quad - 4z_9\, z_{25} - 27z_{10} - 21z_{10}^2 - 24z_{10}\, z_{25} - 6z_{23} - 17z_{25} - 4z_{25}^2 + z_{31} \\
R_{18} &= 2 - 4z_{30}^2 - 8z_{30}\, z_{31} - 4z_{31}^2 + z_{32} \\
R_{19} &= -19z_8 + 8z_8^2 + 8z_8\, z_{19} + 8z_8\, z_{23} - 46z_9 - 24z_9^2 + 24z_9\, z_{19} \\
&\quad + 24z_9\, z_{23} - 2z_{10} + 4z_{19} - 4z_{19}^2 - 8z_{19}\, z_{23} + 14z_{23} - 4z_{23}^2 - z_{25} \\
&\quad + z_{33} \\
R_{20} &= -3 - 6z_8 - 4z_{10} - 2z_{25} - 4z_{33} - 4z_{33}^2 + z_{34} \\
R_{21} &= 8 + 42z_8 + 7z_8^2 + 28z_8\, z_{20} + 48z_9 - 32z_9^2 - 32z_9\, z_{19} \\
&\quad + 64z_9\, z_{20} - 40z_{19}^2 + 32z_{19}\, z_{20} + 14z_{20} - 4z_{20}^2 + z_{35} + 2z_{38} \\
R_{22} &= 1 + 6z_8 - 36z_8^2 - 24z_8\, z_{35} + 2z_{35} - 4z_{35}^2 + z_{36}
\end{aligned}
$$

$$
\begin{aligned}
R_{23} =& -z_9^2 - 4z_9\,z_{36} - 4z_{36}^2 + z_{37}\\
R_{24} =& -6 - 2z_8 - 4z_9 - 14z_{19} - 16z_{19}^2 + z_{38}\\
R_{25} =& \,24 + 21z_9 - 21z_9^2 - 18z_9\,z_{10} - 24z_9\,z_{23} - 24z_9\,z_{38} + 3z_{10}\\
&- 9z_{10}^2 - 16z_{10}\,z_{38} + 3z_{23} - 12z_{23}^2 - 8z_{23}\,z_{38} + 4z_{38} - 4z_{38}^2 + z_{39}\\
&+ 2z_{46}\\
R_{26} =& \,16 - 10z_8 - 20z_8^2 - 16z_8\,z_{10} - 24z_8\,z_{25} - 24z_8\,z_{39} - 12z_{10}\\
&- 16z_{10}\,z_{25} - 16z_{10}\,z_{39} - 14z_{25} - 4z_{25}^2 - 8z_{25}\,z_{39} + 4z_{39} - 4z_{39}^2\\
&+ z_{40}\\
R_{27} =& -z_{10}^2 - 2z_{10}\,z_{11} - z_{11}^2 + z_{41}\\
R_{28} =& \,2 + z_8^2 - 4z_8\,z_{41} + 2z_{10} - 2z_{10}^2 - 8z_{10}\,z_{41} - 2z_{25} + 4z_{41} - 4z_{41}^2\\
&+ z_{42}\\
R_{29} =& -31 - 16z_{10} - 16z_{10}\,z_{23} + 16z_{10}\,z_{25} - 8z_{23} + 12z_{23}^2 - 8z_{23}\,z_{25} + 24z_{25}\\
&- 4z_{25}^2 + z_{43} - 4z_{46}\\
R_{30} =& -4z_{10} - 16z_{25}^2 - 32z_{25}\,z_{41} - 16z_{25}\,z_{42} - 16z_{25}\,z_{43} - 4z_{41} - 16z_{41}^2\\
&- 16z_{41}\,z_{42} - 16z_{41}\,z_{43} - 2z_{42} - 4z_{42}^2 - 8z_{42}\,z_{43} - 4z_{43}^2 + z_{44}\\
R_{31} =& -16z_8^2 + 8z_8\,z_{10} + 32z_8\,z_{25} + 16z_8\,z_{28} + z_{10} - z_{10}^2 - 8z_{10}\,z_{25}\\
&- 4z_{10}\,z_{28} - 16z_{25}^2 - 16z_{25}\,z_{28} - 4z_{28}^2 + z_{45}\\
R_{32} =& -1 - 8z_{10} - 4z_{10}^2 + 8z_{10}\,z_{23} + 8z_{23} - 4z_{23}^2 + z_{46}\\
R_{33} =& -1 - 9z_{10}^2 + 24z_{10}\,z_{23} + 12z_{10}\,z_{46} - 16z_{23}^2 - 16z_{23}\,z_{46} - 4z_{46}^2 + z_{47}\\
R_{34} =& \,1 + z_9 - z_9^2 - 4z_9\,z_{10} - 4z_9\,z_{47} - 4z_{10}^2 - 8z_{10}\,z_{47} - 4z_{47}^2 + z_{48}\\
R_{35} =& -1 - 4z_{48} - 4z_{48}^2 + z_{49}\\
R_{36} =& -1 - z_{10}^2 - 8z_{10}\,z_{48} - 4z_{10}\,z_{49} - 16z_{48}^2 - 16z_{48}\,z_{49} - 4z_{49}^2 + z_{50}\\
R_{37} =& \,2 + z_9 - z_9^2 - 4z_9\,z_{10} - 4z_9\,z_{50} - 4z_{10}^2 - 8z_{10}\,z_{50} - 4z_{50}^2 + z_{51}\\
R_{38} =& -4z_{45}^2 - 8z_{45}\,z_{51} - 4z_{51}^2 + z_{52}\\
R_{39} =& -9 + 7z_8 - z_8^2 - 8z_8\,z_{45} - 4z_8\,z_{52} + 24z_{45} - 16z_{45}^2 - 16z_{45}\,z_{52}\\
&+ 12z_{52} - 4z_{52}^2 + z_{53}\\
R_{40} =& -1 - 4z_{53} - 4z_{53}^2 + z_{54}\\
R_{41} =& -z_{10}^2 - 8z_{10}\,z_{53} - 4z_{10}\,z_{54} - 16z_{53}^2 - 16z_{53}\,z_{54} - 4z_{54}^2 + z_{55}\\
R_{42} =& -z_8 - z_8^2 - 4z_8\,z_{10} - 4z_8\,z_{55} - 4z_{10} - 4z_{10}^2 - 8z_{10}\,z_{55} - 4z_{55}\\
&- 4z_{55}^2 + z_{56}\\
R_{43} =& -10 - 12z_{56} - 4z_{56}^2 + z_{57}\\
R_{44} =& \,2 + 4z_{10} + 16z_{10}\,z_{56} + 8z_{10}\,z_{57} - 4z_{23} - 16z_{23}\,z_{56} - 8z_{23}\,z_{57} - z_{46}\\
&+ 8z_{56} - 16z_{56}^2 - 16z_{56}\,z_{57} + 4z_{57} - 4z_{57}^2 + z_{58}\\
R_{45} =& \,z_9 - z_9^2 + 8z_9\,z_{10} - 8z_9\,z_{23} - 4z_9\,z_{58} - 16z_{10}^2 + 32z_{10}\,z_{23}\\
&+ 16z_{10}\,z_{58} - 16z_{23}^2 - 16z_{23}\,z_{58} - 4z_{58}^2 + z_{59}\\
R_{46} =& -10 - 12z_{51} - 4z_{51}^2 + z_{60}\\
R_{47} =& \,1 - 16z_{51}^2 - 16z_{51}\,z_{59} - 16z_{51}\,z_{60} - 4z_{59}^2 - 8z_{59}\,z_{60} - 4z_{60}^2 + z_{61}\\
R_{48} =& \,z_{10} - z_{10}^2 - 8z_{10}\,z_{59} - 4z_{10}\,z_{61} - 16z_{59}^2 - 16z_{59}\,z_{61} - 4z_{61}^2 + z_{62}
\end{aligned}
$$

$$
\begin{aligned}
R_{49} =& -6z_8 - 3z_8^2 - 18z_8\,z_9 - 16z_9 - 27z_9^2 + 4z_9\,z_{10} + 3z_{10} + 3z_{10}^2 \\
&- 4z_{10}\,z_{19} - 19z_{19} - 8z_{19}^2 + z_{20} + z_{63} - 4z_{76} \\
R_{50} =& \,1 - 8z_8 - 4z_8\,z_{10} + 8z_8\,z_{63} - 16z_9 - 8z_9\,z_{10} + 16z_9\,z_{63} + z_{10} \\
&+ z_{10}^2 + 4z_{10}\,z_{19} + 8z_{19} - 8z_{19}\,z_{63} + 3z_{63} - 4z_{63}^2 + z_{64} \\
R_{51} =& -2 - 12z_8 - 16z_8\,z_{10} + 16z_8\,z_{62} + 16z_8\,z_{64} - 24z_9 - 32z_9\,z_{10} \\
&+ 32z_9\,z_{62} + 32z_9\,z_{64} + 4z_{10} + 4z_{10}^2 + 16z_{10}\,z_{19} + 12z_{19} - 16z_{19}\,z_{62} \\
&- 16z_{19}\,z_{64} - 4z_{62}^2 - 8z_{62}\,z_{64} - 4z_{63} - 2z_{64} - 4z_{64}^2 + z_{65} \\
R_{52} =& -2 - 4z_{65} - 4z_{65}^2 + z_{66} \\
R_{53} =& -z_{13} - 2z_{13}\,z_{228} + z_{67} \\
R_{54} =& \,z_{12} - z_{13} - 2z_{22}\,z_{67} - z_{68} \\
R_{55} =& \,z_{12} - z_{13} - 4z_{24}\,z_{67} - 2z_{27}\,z_{67} - z_{69} \\
R_{56} =& -2z_{29}\,z_{67} + 2z_{44}\,z_{67} - 2z_{67} - z_{70} + z_{75} \\
R_{57} =& \,z_{12} - z_{13} - 4z_{30}\,z_{67} - 2z_{32}\,z_{67} - z_{71} \\
R_{58} =& \,z_{12} - z_{13} - 2z_{34}\,z_{67} + 6z_{67} - z_{72} \\
R_{59} =& \,z_{12} - z_{13} - 4z_{36}\,z_{67} - 2z_{37}\,z_{67} - z_{73} \\
R_{60} =& \,z_{12} - z_{13} - 2z_{40}\,z_{67} - z_{74} \\
R_{61} =& \,z_{12} - z_{13} - 2z_{44}\,z_{67} - 4z_{67} - z_{75} \\
R_{62} =& -z_8\,z_9 + z_8\,z_{10} + z_9 + z_9\,z_{10} - z_{10} - z_{10}^2 + z_{76} \\
R_{63} =& \,1 + 2z_{11} + z_{11}^2 - z_{76}^2 + z_{77} \\
R_{64} =& -z_{77}^2 + z_{78} \\
R_{65} =& -4z_{65}\,z_{67} - 2z_{66}\,z_{67} + 2z_{67} + z_{79} \\
R_{66} =& -z_{12}^2 + 2z_{12}\,z_{13} + 2z_{12}\,z_{79} - z_{13}^2 - 2z_{13}\,z_{79} - z_{79}^2 + z_{80} \\
R_{67} =& -1 - z_9 - 2z_9\,z_{228} + z_{81} \\
R_{68} =& -z_{13}\,z_{81} + z_{83}\,z_{84} + z_{85} \\
R_{69} =& -z_8 - 2z_8\,z_{228} + z_{82} + 2z_{228} \\
R_{70} =& \,1 + 2z_8 + 4z_8\,z_{228} + 4z_9 + 8z_9\,z_{228} - 2z_{19} - 4z_{19}\,z_{228} + z_{83} + 4z_{228} \\
R_{71} =& -z_{12}\,z_{82} + z_{84} \\
R_{72} =& \,z_{12}\,z_{81} + 6z_{13} - 8z_{13}\,z_{81} - 4z_{13}\,z_{82} - z_{13}\,z_{83} - 12z_{13}\,z_{228} + z_{86} \\
R_{73} =& -z_{12}\,z_{68} + z_{13}\,z_{68} + z_{87} \\
R_{74} =& -z_{69}\,z_{87} + z_{88} \\
R_{75} =& -z_{70}\,z_{88} + z_{89} \\
R_{76} =& -z_{71}\,z_{89} + z_{90} \\
R_{77} =& -z_{72}\,z_{90} + z_{91} \\
R_{78} =& -z_{73}\,z_{91} + z_{92} \\
R_{79} =& -z_{74}\,z_{92} + z_{93} \\
R_{80} =& -z_{75}\,z_{93} + z_{94} \\
R_{81} =& -z_{85}\,z_{94} + z_{95} \\
R_{82} =& -z_{86}\,z_{95} + z_{96} \\
R_{83} =& -z_{96}^2 + z_{97}
\end{aligned}
$$

$$
\begin{aligned}
R_{84} &= -z_2 - z_2\,z_{16} + z_{16} + z_{98}\\
R_{85} &= -z_{14}\,z_{98} + z_{99}\\
R_{86} &= -2z_1^2 + 2z_1\,z_{12} + 2z_1\,z_{13} + 2z_1\,z_{99} + 2z_1\,z_{100} - z_{12}^2 - 2z_{12}\,z_{99}\\
&\quad - z_{13}^2 - 2z_{13}\,z_{100} + z_{17}\,z_{102} - z_{78}\,z_{97} - z_{80}\,z_{97} - z_{99}^2 - z_{100}^2\\
R_{87} &= -z_2\,z_{15} + z_{15} - z_{15}\,z_{98} + z_{100}\\
R_{88} &= -z_8 - z_9 - 4z_{19} - 2z_{22} - 4z_{24} - 2z_{27} - 2z_{29} - 4z_{30} - 2z_{32} - 2z_{34}\\
&\quad - 4z_{36} - 2z_{37} - 2z_{40} - 2z_{44} - 4z_{65} - 2z_{66} + z_{101}\\
R_{89} &= -3z_{105} - 6z_{105}\,z_{221} + z_{107}\\
R_{90} &= z_{103}\,z_{212} - z_{106} - z_{107}\,z_{128} - z_{128} - 2z_{128}\,z_{228}\\
R_{91} &= z_{108}\,z_{213} - z_{110}\,z_{131} - z_{111}\\
R_{92} &= z_{112}\,z_{213} - z_{113}\,z_{131} - z_{114}\\
R_{93} &= z_{115}\,z_{213} - z_{116}\,z_{131} - z_{117}\\
R_{94} &= z_{118}\,z_{213} - z_{119}\,z_{131} - z_{120}\\
R_{95} &= z_{121}\,z_{214} - z_{123}\,z_{131} - z_{124}\\
R_{96} &= z_{125}\,z_{214} - z_{126}\,z_{131} - z_{127}\\
R_{97} &= -z_{17}\,z_{128} + z_{131}\\
R_{98} &= -z_{131}\,z_{135} + z_{134}\,z_{212} - z_{136}\\
R_{99} &= -z_{131}\,z_{138} + z_{137}\,z_{213} - z_{139}\\
R_{100} &= z_{140} - 2z_{140}\,z_{142} + z_{143}\\
R_{101} &= -z_{140}\,z_{143} - z_{143} + z_{144}\\
R_{102} &= -1 + z_{143}^2 - z_{144}^2 + z_{224}^2\\
R_{103} &= -z_{141}\,z_{146} - 7z_{146} + z_{147}\\
R_{104} &= 2z_{145} + z_{145}^2 + z_{146}^2 - z_{147}^2\\
R_{105} &= 1 - z_4 - 32z_{141}\,z_{148} + z_{145} + 16z_{146} - z_{147} + 33z_{148}\\
R_{106} &= -12 - 2z_{141} - 4z_{141}\,z_{228} + z_{149} - 24z_{228} + 4z_{228}^2\\
R_{107} &= z_{17} + z_{149}\,z_{150} - z_{183} - z_{184} - 2z_{184}\,z_{228} + z_{185}\\
R_{108} &= -1 + 2z_{141}\,z_{196} - z_{151} + 14z_{196} - z_{196}^2\\
R_{109} &= 3 + z_{151}\,z_{152} - z_{183} - z_{184}\,z_{196} + z_{185} + 6z_{221}\\
R_{110} &= -z_{141}\,z_{154} - 7z_{154} + z_{155}\\
R_{111} &= 2z_{153} + z_{153}^2 + z_{154}^2 - z_{155}^2\\
R_{112} &= -39 + z_{131}\,z_{229} - 8z_{141} - 12z_{141}\,z_{221} - 36z_{221} + 36z_{221}^2\\
R_{113} &= 1 - z_{103} - z_{141}\,z_{154} + z_{153} - 3z_{154} + 6z_{154}\,z_{221}\\
R_{114} &= 7 - z_{131}\,z_{229} + 2z_{141} + z_{156} - 12z_{221}\\
R_{115} &= -1 + z_{104} - z_{145} - 3z_{146} - 6z_{146}\,z_{221} + z_{147} + z_{156}\,z_{157}\\
R_{116} &= 7 + z_{141} - z_{141}\,z_{159} - 7z_{159} + z_{160}\\
R_{117} &= 1 + 2z_{158} + z_{158}^2 - 2z_{159} + z_{159}^2 - z_{160}^2\\
R_{118} &= 4 - z_{108} + z_{141} - z_{141}\,z_{159} + z_{158} - 3z_{159} + 6z_{159}\,z_{221} - 6z_{221}\\
R_{119} &= -3z_{104} - 6z_{104}\,z_{221} + z_{109}\\
R_{120} &= -z_{141}\,z_{162} - 7z_{162} + z_{163}
\end{aligned}
$$

$$
\begin{aligned}
R_{121} &= 2z_{161} + z_{161}^2 + z_{162}^2 - z_{163}^2 \\
R_{122} &= 1 - z_{112} - z_{141}\, z_{162} + z_{161} - 3z_{162} + 6z_{162}\, z_{221} \\
R_{123} &= -z_{141}\, z_{165} - 7z_{165} + z_{166} \\
R_{124} &= 2z_{164} + z_{164}^2 + z_{165}^2 - z_{166}^2 \\
R_{125} &= 1 - z_{115} - z_{141}\, z_{165} + z_{164} - 3z_{165} + 6z_{165}\, z_{221} \\
R_{126} &= -z_{141}\, z_{168} - 7z_{168} + z_{169} \\
R_{127} &= 2z_{167} + z_{167}^2 + z_{168}^2 - z_{169}^2 \\
R_{128} &= 1 - z_{118} - z_{141}\, z_{168} + z_{167} - 3z_{168} + 6z_{168}\, z_{221} \\
R_{129} &= -z_{141}\, z_{171} - 7z_{171} + z_{172} \\
R_{130} &= 2z_{170} + z_{170}^2 + z_{171}^2 - z_{172}^2 \\
R_{131} &= 1 - z_{121} - z_{141}\, z_{171} + z_{170} - 3z_{171} + 6z_{171}\, z_{221} \\
R_{132} &= -z_{141}\, z_{174} - 7z_{174} + z_{175} \\
R_{133} &= 2z_{173} + z_{173}^2 + z_{174}^2 - z_{175}^2 \\
R_{134} &= -1 + z_{122} + z_{156}\, z_{176} - z_{173} - 3z_{174} - 6z_{174}\, z_{221} + z_{175} \\
R_{135} &= -z_{141}\, z_{178} - 7z_{178} + z_{179} \\
R_{136} &= 2z_{177} + z_{177}^2 + z_{178}^2 - z_{179}^2 \\
R_{137} &= 1 - z_{125} - z_{141}\, z_{178} + z_{177} - 3z_{178} + 6z_{178}\, z_{221} \\
R_{138} &= -z_{141}\, z_{181} - 7z_{181} + z_{182} \\
R_{139} &= 2z_{180} + z_{180}^2 + z_{181}^2 - z_{182}^2 \\
R_{140} &= 1 - z_{129}\, z_{131} - z_{130} - z_{141}\, z_{181} + z_{180} - 3z_{181} + 6z_{181}\, z_{221} \\
R_{141} &= 1 - z_{141}\, z_{184} - 7z_{184} + z_{185} \\
R_{142} &= -1 + 2z_{183} + z_{183}^2 + z_{184}^2 - 2z_{185} - z_{185}^2 \\
R_{143} &= z_{128} + z_{156}\, z_{186} - z_{183} - 3z_{184} - 6z_{184}\, z_{221} + z_{185} \\
R_{144} &= -z_{141}\, z_{188} - 7z_{188} + z_{189} \\
R_{145} &= 2z_{187} + z_{187}^2 + z_{188}^2 - z_{189}^2 \\
R_{146} &= 1 - z_{131}\, z_{132} - z_{133} - z_{141}\, z_{188} + z_{187} - 3z_{188} + 6z_{188}\, z_{221} \\
R_{147} &= -z_{141}\, z_{191} - 7z_{191} + z_{192} \\
R_{148} &= 2z_{190} + z_{190}^2 + z_{191}^2 - z_{192}^2 \\
R_{149} &= 1 - z_{134} - z_{141}\, z_{191} + z_{190} - 3z_{191} + 6z_{191}\, z_{221} \\
R_{150} &= -z_{141}\, z_{194} - 7z_{194} + z_{195} \\
R_{151} &= 2z_{193} + z_{193}^2 + z_{194}^2 - z_{195}^2 \\
R_{152} &= 1 - z_{137} - z_{141}\, z_{194} + z_{193} - 3z_{194} + 6z_{194}\, z_{221} \\
R_{153} &= -12 - 2z_0^2 - 2z_0\, z_4 - z_1^2 - z_2^2 - z_4^2 + 2z_8 + 2z_8\, z_9 - z_{10}^2 \\
&\qquad - z_{11}^2 - z_{12}^2 - z_{13}^2 - z_{14}^2 - z_{15}^2 - z_{16}^2 - z_{17}^2 - z_{19} - z_{101}^2 - z_{104}^2 \\
&\qquad - z_{122}^2 - z_{128}^2 + z_{140} - z_{196}^2 - 36z_{221} - 36z_{221}^2 - 4z_{228} - 4z_{228}^2 \\
R_{154} &= z_{196} - z_{227} - 2z_{227}\, z_{228} \\
R_{155} &= -16 + z_0 - z_{197}^2 \\
R_{156} &= -z_{141}\, z_{199} - 7z_{199} + z_{208} \\
R_{157} &= -1 + z_{198}^2 + z_{199}^2 - z_{208}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{158} =\ & -z_{201} - z_{201}\, z_{202} + z_{209} \\
R_{159} =\ & -1 + z_{200}^2 + z_{201}^2 - z_{209}^2 \\
R_{160} =\ & -z_{194}^2 + z_{210} \\
R_{161} =\ & z_{199} - 2z_{204}\, z_{210} - z_{210} \\
R_{162} =\ & -4z_{199} - 8z_{199}\, z_{203} + z_{202} \\
R_{163} =\ & -6 - z_{141} - z_{198}\, z_{205} + z_{202} \\
R_{164} =\ & -1 - z_{193} - z_{198}\, z_{206} + z_{200} \\
R_{165} =\ & -z_{101} - 4z_{194}\, z_{207} + z_{201} \\
R_{166} =\ & -2 + 2z_0^2 + 2z_0\, z_4 - 2z_0\, z_{146} - 2z_0\, z_{154} + z_1^2 - 2z_1\, z_{184} + z_2^2 \\
& - 2z_2\, z_{174} + z_4^2 - 2z_4\, z_{154} + z_8^2 - 2z_8\, z_{159} + z_9^2 - 2z_9\, z_{162} + z_{10}^2 \\
& - 2z_{10}\, z_{165} + z_{11}^2 - 2z_{11}\, z_{168} + z_{12}^2 - 2z_{12}\, z_{171} + z_{13}^2 - 2z_{13}\, z_{178} + z_{14}^2 \\
& - 2z_{14}\, z_{181} + z_{15}^2 - 2z_{15}\, z_{188} + z_{16}^2 - 2z_{16}\, z_{191} - 2z_{101}\, z_{194} - z_{141}\, z_{211} \\
& + z_{145}^2 + z_{153}^2 + z_{158}^2 + 2z_{159} + z_{161}^2 + z_{164}^2 + z_{167}^2 + z_{170}^2 + z_{173}^2 + z_{177}^2 \\
& + z_{180}^2 + z_{183}^2 + z_{187}^2 + z_{190}^2 + z_{193}^2 + 2z_{194}^2 - 6z_{211} - z_{215}^2 - z_{216}^2 - z_{217} \\
& - z_{217}^2 \\
R_{167} =\ & -z_{104}\, z_{212} + z_{128} \\
R_{168} =\ & -z_{109}\, z_{213} + z_{128} \\
R_{169} =\ & -z_{122}\, z_{214} + z_{128} \\
R_{170} =\ & -1 + z_2 - z_3\, z_{222} \\
R_{171} =\ & -2 - z_{101}^2 - z_{146}^2 - z_{154}^2 + 2z_{159} - z_{159}^2 - z_{162}^2 - z_{165}^2 - z_{168}^2 - z_{171}^2 \\
& - z_{174}^2 - z_{178}^2 - z_{181}^2 - z_{184}^2 - z_{188}^2 - z_{191}^2 + z_{194}^2 - z_{215}^2 - z_{216}^2 - z_{217} \\
& - z_{217}^2 \\
R_{172} =\ & -1 - 4z_{106}^2 - 4z_{111}^2 - 4z_{114}^2 - 4z_{117}^2 - 4z_{120}^2 - 4z_{124}^2 - 4z_{127}^2 + z_{128}^2 \\
& - 4z_{130}^2 - 4z_{133}^2 - 4z_{136}^2 - 4z_{139}^2 - z_{218}^2 - z_{219}^2 - z_{220} - z_{220}^2 \\
R_{173} =\ & -1 + z_{222} - 4z_{223} - 4z_{223}^2 \\
R_{174} =\ & 7 + z_{141} - z_{224}^2 - z_{225}^2 - z_{226}^2 \\
R_{175} =\ & -3 - 2z_0 - 2z_4 + 2z_8 - z_8^2 - z_9^2 - z_{10}^2 - z_{11}^2 - z_{12}^2 - z_{13}^2 \\
& - z_{14}^2 - z_{15}^2 - z_{16}^2 + z_{227}
\end{aligned}
$$

B Complete compact residual list

For the companion, the only unknowns are $w_0, \dots, w_{228} \in \mathbb{Z}$. Its equation is

$$Q(\boldsymbol{w}) = \sum_{i=0}^{174} S_i(\boldsymbol{w})^2 = 0.$$

These are exactly the residuals in `equations/compact/equation.json`, with w used as the printed coordinate name. The compact equation has an integer solution if and only if a finite syntactic ZFC refutation exists.

$$\begin{aligned}
S_0 &= w_3 - w_4 - 2w_4\,w_{227}\\
S_1 &= w_0 - w_0\,w_2 - w_2 + w_7\\
S_2 &= -1 - w_2\,w_5 + w_6\,w_7 - 2w_{227}\\
S_3 &= -1 + w_1 - w_6\,w_7\\
S_4 &= 1 + w_4 + w_{16} - w_{18} - 2w_{18}\,w_{227}\\
S_5 &= -1 - w_8^2 - 2w_8\,w_9 - w_9^2 + w_{19}\\
S_6 &= -10w_8 + w_8^2 - 6w_8\,w_9 - 8w_8\,w_{19} - 20w_9 - 11w_9^2 - 16w_9\,w_{19}\\
&\quad - 15w_{19} - 4w_{19}^2 + w_{20}\\
S_7 &= -2 + 2w_8 - w_8^2 - 4w_8\,w_{20} - 4w_{20} - 4w_{20}^2 + w_{21}\\
S_8 &= 1 + 6w_8 - 36w_8^2 - 24w_8\,w_{21} + 2w_{21} - 4w_{21}^2 + w_{22}\\
S_9 &= -2 - w_9 - w_9^2 - 2w_9\,w_{10} - w_{10} - w_{10}^2 + w_{23}\\
S_{10} &= -8 + 7w_8 - w_8^2 + 4w_8\,w_{10} - 4w_8\,w_{23} - 4w_{10} + 4w_{23} + w_{24} - w_{46}\\
S_{11} &= -1 + w_8 - w_8^2 - 2w_8\,w_{10} + w_{10} - w_{10}^2 + w_{25}\\
S_{12} &= 23 + 16w_8 - 8w_8\,w_9 - 8w_8\,w_{10} - 16w_8\,w_{19} - 8w_8\,w_{25} + 19w_9\\
&\quad + 9w_9^2 - 36w_9\,w_{10} - 12w_9\,w_{25} + 16w_{10} + 8w_{10}^2 - 24w_{10}\,w_{19} - 8w_{19}\,w_{25}\\
&\quad - w_{20} + w_{26} - w_{31}\\
S_{13} &= -4w_{24}^2 - 8w_{24}\,w_{26} - 4w_{26}^2 + w_{27}\\
S_{14} &= -2 - 8w_8 + 8w_8\,w_{23} + 8w_8\,w_{25} - 8w_9 + 8w_9\,w_{23} + 8w_9\,w_{25} - 4w_{19}\\
&\quad + 8w_{23} - 4w_{23}^2 - 8w_{23}\,w_{25} + 8w_{25} - 4w_{25}^2 + w_{28}\\
S_{15} &= 6 + 4w_8 + 32w_8\,w_9 + 16w_8\,w_{19} + 16w_8\,w_{25} + 8w_8\,w_{28} + 24w_9\\
&\quad - 32w_9\,w_{25} - 16w_9\,w_{28} + 12w_{19} - 16w_{19}\,w_{25} - 8w_{19}\,w_{28} - w_{20} + 8w_{25}\\
&\quad - 16w_{25}^2 - 16w_{25}\,w_{28} + 4w_{28} - 4w_{28}^2 + w_{29}\\
S_{16} &= 22 - 5w_8 - w_8^2 - 8w_8\,w_9 - 12w_8\,w_{10} - 4w_8\,w_{23} - 8w_9 - 16w_9\,w_{10}\\
&\quad - 16w_9\,w_{23} - 20w_{10} - 20w_{10}^2 - 24w_{10}\,w_{23} - 28w_{23} - 4w_{23}^2 + w_{30}\\
S_{17} &= 19 - 25w_8 - 7w_8^2 - 8w_8\,w_9 - 30w_8\,w_{10} - 16w_8\,w_{25} + 3w_9 + 5w_9^2\\
&\quad - 4w_9\,w_{25} - 27w_{10} - 21w_{10}^2 - 24w_{10}\,w_{25} - 6w_{23} - 17w_{25} - 4w_{25}^2 + w_{31}\\
S_{18} &= 2 - 4w_{30}^2 - 8w_{30}\,w_{31} - 4w_{31}^2 + w_{32}\\
S_{19} &= -19w_8 + 8w_8^2 + 8w_8\,w_{19} + 8w_8\,w_{23} - 46w_9 - 24w_9^2 + 24w_9\,w_{19}\\
&\quad + 24w_9\,w_{23} - 2w_{10} + 4w_{19} - 4w_{19}^2 - 8w_{19}\,w_{23} + 14w_{23} - 4w_{23}^2 - w_{25}\\
&\quad + w_{33}\\
S_{20} &= -3 - 6w_8 - 4w_{10} - 2w_{25} - 4w_{33} - 4w_{33}^2 + w_{34}
\end{aligned}$$

$$
\begin{aligned}
S_{21} &= 8 + 42w_8 + 7w_8^2 + 28w_8\,w_{20} + 48w_9 - 32w_9^2 - 32w_9\,w_{19} \\
&\quad + 64w_9\,w_{20} - 40w_{19}^2 + 32w_{19}\,w_{20} + 14w_{20} - 4w_{20}^2 + w_{35} + 2w_{38} \\
S_{22} &= 1 + 6w_8 - 36w_8^2 - 24w_8\,w_{35} + 2w_{35} - 4w_{35}^2 + w_{36} \\
S_{23} &= -w_9^2 - 4w_9\,w_{36} - 4w_{36}^2 + w_{37} \\
S_{24} &= -6 - 2w_8 - 4w_9 - 14w_{19} - 16w_{19}^2 + w_{38} \\
S_{25} &= 24 + 21w_9 - 21w_9^2 - 18w_9\,w_{10} - 24w_9\,w_{23} - 24w_9\,w_{38} + 3w_{10} \\
&\quad - 9w_{10}^2 - 16w_{10}\,w_{38} + 3w_{23} - 12w_{23}^2 - 8w_{23}\,w_{38} + 4w_{38} - 4w_{38}^2 + w_{39} \\
&\quad + 2w_{46} \\
S_{26} &= 16 - 10w_8 - 20w_8^2 - 16w_8\,w_{10} - 24w_8\,w_{25} - 24w_8\,w_{39} - 12w_{10} \\
&\quad - 16w_{10}\,w_{25} - 16w_{10}\,w_{39} - 14w_{25} - 4w_{25}^2 - 8w_{25}\,w_{39} + 4w_{39} - 4w_{39}^2 \\
&\quad + w_{40} \\
S_{27} &= -w_{10}^2 - 2w_{10}\,w_{11} - w_{11}^2 + w_{41} \\
S_{28} &= 2 + w_8^2 - 4w_8\,w_{41} + 2w_{10} - 2w_{10}^2 - 8w_{10}\,w_{41} - 2w_{25} + 4w_{41} - 4w_{41}^2 \\
&\quad + w_{42} \\
S_{29} &= -31 - 16w_{10} - 16w_{10}\,w_{23} + 16w_{10}\,w_{25} - 8w_{23} + 12w_{23}^2 - 8w_{23}\,w_{25} + 24w_{25} \\
&\quad - 4w_{25}^2 + w_{43} - 4w_{46} \\
S_{30} &= -4w_{10} - 16w_{25}^2 - 32w_{25}\,w_{41} - 16w_{25}\,w_{42} - 16w_{25}\,w_{43} - 4w_{41} - 16w_{41}^2 \\
&\quad - 16w_{41}\,w_{42} - 16w_{41}\,w_{43} - 2w_{42} - 4w_{42}^2 - 8w_{42}\,w_{43} - 4w_{43}^2 + w_{44} \\
S_{31} &= -16w_8^2 + 8w_8\,w_{10} + 32w_8\,w_{25} + 16w_8\,w_{28} + w_{10} - w_{10}^2 - 8w_{10}\,w_{25} \\
&\quad - 4w_{10}\,w_{28} - 16w_{25}^2 - 16w_{25}\,w_{28} - 4w_{28}^2 + w_{45} \\
S_{32} &= -1 - 8w_{10} - 4w_{10}^2 + 8w_{10}\,w_{23} + 8w_{23} - 4w_{23}^2 + w_{46} \\
S_{33} &= -1 - 9w_{10}^2 + 24w_{10}\,w_{23} + 12w_{10}\,w_{46} - 16w_{23}^2 - 16w_{23}\,w_{46} - 4w_{46}^2 + w_{47} \\
S_{34} &= 1 + w_9 - w_9^2 - 4w_9\,w_{10} - 4w_9\,w_{47} - 4w_{10}^2 - 8w_{10}\,w_{47} - 4w_{47}^2 + w_{48} \\
S_{35} &= -1 - 4w_{48} - 4w_{48}^2 + w_{49} \\
S_{36} &= -1 - w_{10}^2 - 8w_{10}\,w_{48} - 4w_{10}\,w_{49} - 16w_{48}^2 - 16w_{48}\,w_{49} - 4w_{49}^2 + w_{50} \\
S_{37} &= 2 + w_9 - w_9^2 - 4w_9\,w_{10} - 4w_9\,w_{50} - 4w_{10}^2 - 8w_{10}\,w_{50} - 4w_{50}^2 + w_{51} \\
S_{38} &= -4w_{45}^2 - 8w_{45}\,w_{51} - 4w_{51}^2 + w_{52} \\
S_{39} &= -9 + 7w_8 - w_8^2 - 8w_8\,w_{45} - 4w_8\,w_{52} + 24w_{45} - 16w_{45}^2 - 16w_{45}\,w_{52} \\
&\quad + 12w_{52} - 4w_{52}^2 + w_{53} \\
S_{40} &= -1 - 4w_{53} - 4w_{53}^2 + w_{54} \\
S_{41} &= -w_{10}^2 - 8w_{10}\,w_{53} - 4w_{10}\,w_{54} - 16w_{53}^2 - 16w_{53}\,w_{54} - 4w_{54}^2 + w_{55} \\
S_{42} &= -w_8 - w_8^2 - 4w_8\,w_{10} - 4w_8\,w_{55} - 4w_{10} - 4w_{10}^2 - 8w_{10}\,w_{55} - 4w_{55} \\
&\quad - 4w_{55}^2 + w_{56} \\
S_{43} &= -10 - 12w_{56} - 4w_{56}^2 + w_{57} \\
S_{44} &= 2 + 4w_{10} + 16w_{10}\,w_{56} + 8w_{10}\,w_{57} - 4w_{23} - 16w_{23}\,w_{56} - 8w_{23}\,w_{57} - w_{46} \\
&\quad + 8w_{56} - 16w_{56}^2 - 16w_{56}\,w_{57} + 4w_{57} - 4w_{57}^2 + w_{58} \\
S_{45} &= w_9 - w_9^2 + 8w_9\,w_{10} - 8w_9\,w_{23} - 4w_9\,w_{58} - 16w_{10}^2 + 32w_{10}\,w_{23} \\
&\quad + 16w_{10}\,w_{58} - 16w_{23}^2 - 16w_{23}\,w_{58} - 4w_{58}^2 + w_{59} \\
S_{46} &= -10 - 12w_{51} - 4w_{51}^2 + w_{60}
\end{aligned}
$$

$$
\begin{aligned}
S_{47} &= 1 - 16w_{51}^2 - 16w_{51}\,w_{59} - 16w_{51}\,w_{60} - 4w_{59}^2 - 8w_{59}\,w_{60} - 4w_{60}^2 + w_{61}\\
S_{48} &= w_{10} - w_{10}^2 - 8w_{10}\,w_{59} - 4w_{10}\,w_{61} - 16w_{59}^2 - 16w_{59}\,w_{61} - 4w_{61}^2 + w_{62}\\
S_{49} &= -6w_8 - 3w_8^2 - 18w_8\,w_9 - 16w_9 - 27w_9^2 + 4w_9\,w_{10} + 3w_{10} + 3w_{10}^2\\
&\quad - 4w_{10}\,w_{19} - 19w_{19} - 8w_{19}^2 + w_{20} + w_{63} - 4w_{76}\\
S_{50} &= 1 - 8w_8 - 4w_8\,w_{10} + 8w_8\,w_{63} - 16w_9 - 8w_9\,w_{10} + 16w_9\,w_{63} + w_{10}\\
&\quad + w_{10}^2 + 4w_{10}\,w_{19} + 8w_{19} - 8w_{19}\,w_{63} + 3w_{63} - 4w_{63}^2 + w_{64}\\
S_{51} &= -2 - 12w_8 - 16w_8\,w_{10} + 16w_8\,w_{62} + 16w_8\,w_{64} - 24w_9 - 32w_9\,w_{10}\\
&\quad + 32w_9\,w_{62} + 32w_9\,w_{64} + 4w_{10} + 4w_{10}^2 + 16w_{10}\,w_{19} + 12w_{19} - 16w_{19}\,w_{62}\\
&\quad - 16w_{19}\,w_{64} - 4w_{62}^2 - 8w_{62}\,w_{64} - 4w_{63} - 2w_{64} - 4w_{64}^2 + w_{65}\\
S_{52} &= -2 - 4w_{65} - 4w_{65}^2 + w_{66}\\
S_{53} &= -w_{13} - 2w_{13}\,w_{227} + w_{67}\\
S_{54} &= w_{12} - w_{13} - 2w_{22}\,w_{67} - w_{68}\\
S_{55} &= w_{12} - w_{13} - 4w_{24}\,w_{67} - 2w_{27}\,w_{67} - w_{69}\\
S_{56} &= -2w_{29}\,w_{67} + 2w_{44}\,w_{67} - 2w_{67} - w_{70} + w_{75}\\
S_{57} &= w_{12} - w_{13} - 4w_{30}\,w_{67} - 2w_{32}\,w_{67} - w_{71}\\
S_{58} &= w_{12} - w_{13} - 2w_{34}\,w_{67} + 6w_{67} - w_{72}\\
S_{59} &= w_{12} - w_{13} - 4w_{36}\,w_{67} - 2w_{37}\,w_{67} - w_{73}\\
S_{60} &= w_{12} - w_{13} - 2w_{40}\,w_{67} - w_{74}\\
S_{61} &= w_{12} - w_{13} - 2w_{44}\,w_{67} - 4w_{67} - w_{75}\\
S_{62} &= -w_8\,w_9 + w_8\,w_{10} + w_9 + w_9\,w_{10} - w_{10} - w_{10}^2 + w_{76}\\
S_{63} &= 1 + 2w_{11} + w_{11}^2 - w_{76}^2 + w_{77}\\
S_{64} &= -w_{77}^2 + w_{78}\\
S_{65} &= -4w_{65}\,w_{67} - 2w_{66}\,w_{67} + 2w_{67} + w_{79}\\
S_{66} &= -w_{12}^2 + 2w_{12}\,w_{13} + 2w_{12}\,w_{79} - w_{13}^2 - 2w_{13}\,w_{79} - w_{79}^2 + w_{80}\\
S_{67} &= -1 - w_9 - 2w_9\,w_{227} + w_{81}\\
S_{68} &= -w_{13}\,w_{81} + w_{83}\,w_{84} + w_{85}\\
S_{69} &= -w_8 - 2w_8\,w_{227} + w_{82} + 2w_{227}\\
S_{70} &= 1 + 2w_8 + 4w_8\,w_{227} + 4w_9 + 8w_9\,w_{227} - 2w_{19} - 4w_{19}\,w_{227} + w_{83} + 4w_{227}\\
S_{71} &= -w_{12}\,w_{82} + w_{84}\\
S_{72} &= w_{12}\,w_{81} + 6w_{13} - 8w_{13}\,w_{81} - 4w_{13}\,w_{82} - w_{13}\,w_{83} - 12w_{13}\,w_{227} + w_{86}\\
S_{73} &= -w_{12}\,w_{68} + w_{13}\,w_{68} + w_{87}\\
S_{74} &= -w_{69}\,w_{87} + w_{88}\\
S_{75} &= -w_{70}\,w_{88} + w_{89}\\
S_{76} &= -w_{71}\,w_{89} + w_{90}\\
S_{77} &= -w_{72}\,w_{90} + w_{91}\\
S_{78} &= -w_{73}\,w_{91} + w_{92}\\
S_{79} &= -w_{74}\,w_{92} + w_{93}\\
S_{80} &= -w_{75}\,w_{93} + w_{94}\\
S_{81} &= -w_{85}\,w_{94} + w_{95}
\end{aligned}
$$

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\begin{aligned}
S_{82} &= -w_{86}\,w_{95} + w_{96}\\
S_{83} &= -w_{96}^2 + w_{97}\\
S_{84} &= -w_2 - w_2\,w_{16} + w_{16} + w_{98}\\
S_{85} &= -w_{14}\,w_{98} + w_{99}\\
S_{86} &= -2w_1^2 + 2w_1\,w_{12} + 2w_1\,w_{13} + 2w_1\,w_{99} + 2w_1\,w_{100} - w_{12}^2 - 2w_{12}\,w_{99}\\
&\quad - w_{13}^2 - 2w_{13}\,w_{100} + w_{17}\,w_{101} - w_{78}\,w_{97} - w_{80}\,w_{97} - w_{99}^2 - w_{100}^2\\
S_{87} &= -w_2\,w_{15} + w_{15} - w_{15}\,w_{98} + w_{100}\\
S_{88} &= -3w_{104} - 6w_{104}\,w_{220} + w_{106}\\
S_{89} &= w_{102}\,w_{211} - w_{105} - w_{106}\,w_{127} - w_{127} - 2w_{127}\,w_{227}\\
S_{90} &= w_{107}\,w_{212} - w_{109}\,w_{130} - w_{110}\\
S_{91} &= w_{111}\,w_{212} - w_{112}\,w_{130} - w_{113}\\
S_{92} &= w_{114}\,w_{212} - w_{115}\,w_{130} - w_{116}\\
S_{93} &= w_{117}\,w_{212} - w_{118}\,w_{130} - w_{119}\\
S_{94} &= w_{120}\,w_{213} - w_{122}\,w_{130} - w_{123}\\
S_{95} &= w_{124}\,w_{213} - w_{125}\,w_{130} - w_{126}\\
S_{96} &= -w_{17}\,w_{127} + w_{130}\\
S_{97} &= -w_{130}\,w_{134} + w_{133}\,w_{211} - w_{135}\\
S_{98} &= -w_{130}\,w_{137} + w_{136}\,w_{212} - w_{138}\\
S_{99} &= w_{139} - 2w_{139}\,w_{141} + w_{142}\\
S_{100} &= -w_{139}\,w_{142} - w_{142} + w_{143}\\
S_{101} &= -1 + w_{142}^2 - w_{143}^2 + w_{223}^2\\
S_{102} &= -w_{140}\,w_{145} - 7w_{145} + w_{146}\\
S_{103} &= 2w_{144} + w_{144}^2 + w_{145}^2 - w_{146}^2\\
S_{104} &= 1 - w_4 - 32w_{140}\,w_{147} + w_{144} + 16w_{145} - w_{146} + 33w_{147}\\
S_{105} &= -12 - 2w_{140} - 4w_{140}\,w_{227} + w_{148} - 24w_{227} + 4w_{227}^2\\
S_{106} &= w_{17} + w_{148}\,w_{149} - w_{182} - w_{183} - 2w_{183}\,w_{227} + w_{184}\\
S_{107} &= -1 + 2w_{140}\,w_{195} - w_{150} + 14w_{195} - w_{195}^2\\
S_{108} &= 3 + w_{150}\,w_{151} - w_{182} - w_{183}\,w_{195} + w_{184} + 6w_{220}\\
S_{109} &= -w_{140}\,w_{153} - 7w_{153} + w_{154}\\
S_{110} &= 2w_{152} + w_{152}^2 + w_{153}^2 - w_{154}^2\\
S_{111} &= -39 + w_{130}\,w_{228} - 8w_{140} - 12w_{140}\,w_{220} - 36w_{220} + 36w_{220}^2\\
S_{112} &= 1 - w_{102} - w_{140}\,w_{153} + w_{152} - 3w_{153} + 6w_{153}\,w_{220}\\
S_{113} &= 7 - w_{130}\,w_{228} + 2w_{140} + w_{155} - 12w_{220}\\
S_{114} &= -1 + w_{103} - w_{144} - 3w_{145} - 6w_{145}\,w_{220} + w_{146} + w_{155}\,w_{156}\\
S_{115} &= 7 + w_{140} - w_{140}\,w_{158} - 7w_{158} + w_{159}\\
S_{116} &= 1 + 2w_{157} + w_{157}^2 - 2w_{158} + w_{158}^2 - w_{159}^2\\
S_{117} &= 4 - w_{107} + w_{140} - w_{140}\,w_{158} + w_{157} - 3w_{158} + 6w_{158}\,w_{220} - 6w_{220}\\
S_{118} &= -3w_{103} - 6w_{103}\,w_{220} + w_{108}\\
S_{119} &= -w_{140}\,w_{161} - 7w_{161} + w_{162}
\end{aligned}
$$

$$
\begin{aligned}
S_{120} &= 2w_{160} + w_{160}^2 + w_{161}^2 - w_{162}^2 \\
S_{121} &= 1 - w_{111} - w_{140}\,w_{161} + w_{160} - 3w_{161} + 6w_{161}\,w_{220} \\
S_{122} &= -w_{140}\,w_{164} - 7w_{164} + w_{165} \\
S_{123} &= 2w_{163} + w_{163}^2 + w_{164}^2 - w_{165}^2 \\
S_{124} &= 1 - w_{114} - w_{140}\,w_{164} + w_{163} - 3w_{164} + 6w_{164}\,w_{220} \\
S_{125} &= -w_{140}\,w_{167} - 7w_{167} + w_{168} \\
S_{126} &= 2w_{166} + w_{166}^2 + w_{167}^2 - w_{168}^2 \\
S_{127} &= 1 - w_{117} - w_{140}\,w_{167} + w_{166} - 3w_{167} + 6w_{167}\,w_{220} \\
S_{128} &= -w_{140}\,w_{170} - 7w_{170} + w_{171} \\
S_{129} &= 2w_{169} + w_{169}^2 + w_{170}^2 - w_{171}^2 \\
S_{130} &= 1 - w_{120} - w_{140}\,w_{170} + w_{169} - 3w_{170} + 6w_{170}\,w_{220} \\
S_{131} &= -w_{140}\,w_{173} - 7w_{173} + w_{174} \\
S_{132} &= 2w_{172} + w_{172}^2 + w_{173}^2 - w_{174}^2 \\
S_{133} &= -1 + w_{121} + w_{155}\,w_{175} - w_{172} - 3w_{173} - 6w_{173}\,w_{220} + w_{174} \\
S_{134} &= -w_{140}\,w_{177} - 7w_{177} + w_{178} \\
S_{135} &= 2w_{176} + w_{176}^2 + w_{177}^2 - w_{178}^2 \\
S_{136} &= 1 - w_{124} - w_{140}\,w_{177} + w_{176} - 3w_{177} + 6w_{177}\,w_{220} \\
S_{137} &= -w_{140}\,w_{180} - 7w_{180} + w_{181} \\
S_{138} &= 2w_{179} + w_{179}^2 + w_{180}^2 - w_{181}^2 \\
S_{139} &= 1 - w_{128}\,w_{130} - w_{129} - w_{140}\,w_{180} + w_{179} - 3w_{180} + 6w_{180}\,w_{220} \\
S_{140} &= 1 - w_{140}\,w_{183} - 7w_{183} + w_{184} \\
S_{141} &= -1 + 2w_{182} + w_{182}^2 + w_{183}^2 - 2w_{184} - w_{184}^2 \\
S_{142} &= w_{127} + w_{155}\,w_{185} - w_{182} - 3w_{183} - 6w_{183}\,w_{220} + w_{184} \\
S_{143} &= -w_{140}\,w_{187} - 7w_{187} + w_{188} \\
S_{144} &= 2w_{186} + w_{186}^2 + w_{187}^2 - w_{188}^2 \\
S_{145} &= 1 - w_{130}\,w_{131} - w_{132} - w_{140}\,w_{187} + w_{186} - 3w_{187} + 6w_{187}\,w_{220} \\
S_{146} &= -w_{140}\,w_{190} - 7w_{190} + w_{191} \\
S_{147} &= 2w_{189} + w_{189}^2 + w_{190}^2 - w_{191}^2 \\
S_{148} &= 1 - w_{133} - w_{140}\,w_{190} + w_{189} - 3w_{190} + 6w_{190}\,w_{220} \\
S_{149} &= -w_{140}\,w_{193} - 7w_{193} + w_{194} \\
S_{150} &= 2w_{192} + w_{192}^2 + w_{193}^2 - w_{194}^2 \\
S_{151} &= 1 - w_{136} - w_{140}\,w_{193} + w_{192} - 3w_{193} + 6w_{193}\,w_{220} \\
S_{152} &= -10 - 2w_0^2 - 2w_0\,w_4 - w_1^2 - w_2^2 - w_4^2 + 2w_8 + 2w_8\,w_9 - w_{10}^2 \\
&\quad - w_{11}^2 - w_{12}^2 - w_{13}^2 - w_{14}^2 - w_{15}^2 - w_{16}^2 - w_{17}^2 - w_{19} - w_{103}^2 - w_{121}^2 \\
&\quad - w_{127}^2 + w_{139} + w_{145}^2 + w_{153}^2 - 2w_{158} + w_{158}^2 + w_{161}^2 + w_{164}^2 + w_{167}^2 + w_{170}^2 \\
&\quad + w_{173}^2 + w_{177}^2 + w_{180}^2 + w_{183}^2 + w_{187}^2 + w_{190}^2 - w_{193}^2 - w_{195}^2 + w_{214}^2 + w_{215}^2 \\
&\quad + w_{216} + w_{216}^2 - 36w_{220} - 36w_{220}^2 - 4w_{227} - 4w_{227}^2 \\
S_{153} &= w_{195} - w_{226} - 2w_{226}\,w_{227} \\
S_{154} &= -16 + w_0 - w_{196}^2
\end{aligned}
$$

$$
\begin{aligned}
S_{155} =& -w_{140}\,w_{198} - 7w_{198} + w_{207}\\
S_{156} =& -1 + w_{197}^2 + w_{198}^2 - w_{207}^2\\
S_{157} =& -w_{200} - w_{200}\,w_{201} + w_{208}\\
S_{158} =& -1 + w_{199}^2 + w_{200}^2 - w_{208}^2\\
S_{159} =& -w_{193}^2 + w_{209}\\
S_{160} =& \,w_{198} - 2w_{203}\,w_{209} - w_{209}\\
S_{161} =& -4w_{198} - 8w_{198}\,w_{202} + w_{201}\\
S_{162} =& -6 - w_{140} - w_{197}\,w_{204} + w_{201}\\
S_{163} =& -1 - w_{192} - w_{197}\,w_{205} + w_{199}\\
S_{164} =& -w_8 - w_9 - 4w_{19} - 2w_{22} - 4w_{24} - 2w_{27} - 2w_{29} - 4w_{30} - 2w_{32} - 2w_{34}\\
&- 4w_{36} - 2w_{37} - 2w_{40} - 2w_{44} - 4w_{65} - 2w_{66} - 4w_{193}\,w_{206} + w_{200}\\
S_{165} =& -2 + 2w_0^2 + 2w_0\,w_4 - 2w_0\,w_{145} - 2w_0\,w_{153} + w_1^2 - 2w_1\,w_{183} + w_2^2\\
&- 2w_2\,w_{173} + w_4^2 - 2w_4\,w_{153} + w_8^2 - 2w_8\,w_{158} - 2w_8\,w_{193} + w_9^2\\
&- 2w_9\,w_{161} - 2w_9\,w_{193} + w_{10}^2 - 2w_{10}\,w_{164} + w_{11}^2 - 2w_{11}\,w_{167} + w_{12}^2\\
&- 2w_{12}\,w_{170} + w_{13}^2 - 2w_{13}\,w_{177} + w_{14}^2 - 2w_{14}\,w_{180} + w_{15}^2 - 2w_{15}\,w_{187} + w_{16}^2\\
&- 2w_{16}\,w_{190} - 8w_{19}\,w_{193} - 4w_{22}\,w_{193} - 8w_{24}\,w_{193} - 4w_{27}\,w_{193} - 4w_{29}\,w_{193}\\
&- 8w_{30}\,w_{193} - 4w_{32}\,w_{193} - 4w_{34}\,w_{193} - 8w_{36}\,w_{193} - 4w_{37}\,w_{193} - 4w_{40}\,w_{193}\\
&- 4w_{44}\,w_{193} - 8w_{65}\,w_{193} - 4w_{66}\,w_{193} - w_{140}\,w_{210} + w_{144}^2 + w_{152}^2 + w_{157}^2 + 2w_{158}\\
&+ w_{160}^2 + w_{163}^2 + w_{166}^2 + w_{169}^2 + w_{172}^2 + w_{176}^2 + w_{179}^2 + w_{182}^2 + w_{186}^2 + w_{189}^2\\
&+ w_{192}^2 + 2w_{193}^2 - 6w_{210} - w_{214}^2 - w_{215}^2 - w_{216} - w_{216}^2\\
S_{166} =& -w_{103}\,w_{211} + w_{127}\\
S_{167} =& -w_{108}\,w_{212} + w_{127}\\
S_{168} =& -w_{121}\,w_{213} + w_{127}\\
S_{169} =& -1 + w_2 - w_3\,w_{221}
\end{aligned}
$$

$$
\begin{aligned}
S_{170} = \; & -1 - 8w_8\,w_{19} - 4w_8\,w_{22} - 8w_8\,w_{24} - 4w_8\,w_{27} - 4w_8\,w_{29} - 8w_8\,w_{30} \\
& - 4w_8\,w_{32} - 4w_8\,w_{34} - 8w_8\,w_{36} - 4w_8\,w_{37} - 4w_8\,w_{40} - 4w_8\,w_{44} \\
& - 8w_8\,w_{65} - 4w_8\,w_{66} - 8w_9\,w_{19} - 4w_9\,w_{22} - 8w_9\,w_{24} - 4w_9\,w_{27} \\
& - 4w_9\,w_{29} - 8w_9\,w_{30} - 4w_9\,w_{32} - 4w_9\,w_{34} - 8w_9\,w_{36} - 4w_9\,w_{37} \\
& - 4w_9\,w_{40} - 4w_9\,w_{44} - 8w_9\,w_{65} - 4w_9\,w_{66} - w_{19} - 16w_{19}^2 - 16w_{19}\,w_{22} \\
& - 32w_{19}\,w_{24} - 16w_{19}\,w_{27} - 16w_{19}\,w_{29} - 32w_{19}\,w_{30} - 16w_{19}\,w_{32} - 16w_{19}\,w_{34} \\
& - 32w_{19}\,w_{36} - 16w_{19}\,w_{37} - 16w_{19}\,w_{40} - 16w_{19}\,w_{44} - 32w_{19}\,w_{65} - 16w_{19}\,w_{66} \\
& - 4w_{22}^2 - 16w_{22}\,w_{24} - 8w_{22}\,w_{27} - 8w_{22}\,w_{29} - 16w_{22}\,w_{30} - 8w_{22}\,w_{32} \\
& - 8w_{22}\,w_{34} - 16w_{22}\,w_{36} - 8w_{22}\,w_{37} - 8w_{22}\,w_{40} - 8w_{22}\,w_{44} - 16w_{22}\,w_{65} \\
& - 8w_{22}\,w_{66} - 16w_{24}^2 - 16w_{24}\,w_{27} - 16w_{24}\,w_{29} - 32w_{24}\,w_{30} - 16w_{24}\,w_{32} \\
& - 16w_{24}\,w_{34} - 32w_{24}\,w_{36} - 16w_{24}\,w_{37} - 16w_{24}\,w_{40} - 16w_{24}\,w_{44} - 32w_{24}\,w_{65} \\
& - 16w_{24}\,w_{66} - 4w_{27}^2 - 8w_{27}\,w_{29} - 16w_{27}\,w_{30} - 8w_{27}\,w_{32} - 8w_{27}\,w_{34} \\
& - 16w_{27}\,w_{36} - 8w_{27}\,w_{37} - 8w_{27}\,w_{40} - 8w_{27}\,w_{44} - 16w_{27}\,w_{65} - 8w_{27}\,w_{66} \\
& - 4w_{29}^2 - 16w_{29}\,w_{30} - 8w_{29}\,w_{32} - 8w_{29}\,w_{34} - 16w_{29}\,w_{36} - 8w_{29}\,w_{37} \\
& - 8w_{29}\,w_{40} - 8w_{29}\,w_{44} - 16w_{29}\,w_{65} - 8w_{29}\,w_{66} - 16w_{30}^2 - 16w_{30}\,w_{32} \\
& - 16w_{30}\,w_{34} - 32w_{30}\,w_{36} - 16w_{30}\,w_{37} - 16w_{30}\,w_{40} - 16w_{30}\,w_{44} - 32w_{30}\,w_{65} \\
& - 16w_{30}\,w_{66} - 4w_{32}^2 - 8w_{32}\,w_{34} - 16w_{32}\,w_{36} - 8w_{32}\,w_{37} - 8w_{32}\,w_{40} \\
& - 8w_{32}\,w_{44} - 16w_{32}\,w_{65} - 8w_{32}\,w_{66} - 4w_{34}^2 - 16w_{34}\,w_{36} - 8w_{34}\,w_{37} \\
& - 8w_{34}\,w_{40} - 8w_{34}\,w_{44} - 16w_{34}\,w_{65} - 8w_{34}\,w_{66} - 16w_{36}^2 - 16w_{36}\,w_{37} \\
& - 16w_{36}\,w_{40} - 16w_{36}\,w_{44} - 32w_{36}\,w_{65} - 16w_{36}\,w_{66} - 4w_{37}^2 - 8w_{37}\,w_{40} \\
& - 8w_{37}\,w_{44} - 16w_{37}\,w_{65} - 8w_{37}\,w_{66} - 4w_{40}^2 - 8w_{40}\,w_{44} - 16w_{40}\,w_{65} \\
& - 8w_{40}\,w_{66} - 4w_{44}^2 - 16w_{44}\,w_{65} - 8w_{44}\,w_{66} - 16w_{65}^2 - 16w_{65}\,w_{66} - 4w_{66}^2 \\
& - w_{145}^2 - w_{153}^2 + 2w_{158} - w_{158}^2 - w_{161}^2 - w_{164}^2 - w_{167}^2 - w_{170}^2 - w_{173}^2 - w_{177}^2 \\
& - w_{180}^2 - w_{183}^2 - w_{187}^2 - w_{190}^2 + w_{193}^2 - w_{214}^2 - w_{215}^2 - w_{216} - w_{216}^2 \\
S_{171} = \; & -1 - 4w_{105}^2 - 4w_{110}^2 - 4w_{113}^2 - 4w_{116}^2 - 4w_{119}^2 - 4w_{123}^2 - 4w_{126}^2 + w_{127}^2 \\
& - 4w_{129}^2 - 4w_{132}^2 - 4w_{135}^2 - 4w_{138}^2 - w_{217}^2 - w_{218}^2 - w_{219} - w_{219}^2 \\
S_{172} = \; & -1 + w_{221} - 4w_{222} - 4w_{222}^2 \\
S_{173} = \; & 7 + w_{140} - w_{223}^2 - w_{224}^2 - w_{225}^2 \\
S_{174} = \; & -3 - 2w_0 - 2w_4 + 2w_8 - w_8^2 - w_9^2 - w_{10}^2 - w_{11}^2 - w_{12}^2 - w_{13}^2 \\
& - w_{14}^2 - w_{15}^2 - w_{16}^2 + w_{226}
\end{aligned}
$$